

Robot Perception in Visually Challenging Environments

Akin Mert Gümüs

Master Thesis



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By
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Abstract

Autonomous platforms increasingly need to operate where GNSS is unavailable and visible-light cameras degrade or fail—in darkness, smoke, or dust. Thermal cameras sense emitted radiation rather than reflected light and remain usable in these conditions, making thermal-inertial odometry (TIO) an attractive alternative to conventional visual-inertial odometry (VIO). Classical filter-based VIO, and in particular the Multi-State Constraint Kalman Filter (MSCKF), is well established on visible-spectrum data, but has not been systematically characterised on thermal imagery, where low contrast, radiometric instability, and high-bit-depth raw output challenge the assumptions of standard feature-based front-ends. This thesis implements a monocular MSCKF-based TIO pipeline and evaluates it under a common, reproducible protocol across four datasets spanning visible and thermal modalities, indoor and outdoor scenes, and aerial and ground platforms. Two feature front-ends (KLT and ORB) and two thermal preprocessing orderings (contrast enhancement and 8-bit normalisation) are compared, and trajectories are evaluated with a 4-degree-of-freedom alignment that matches the unobservable directions of visual-inertial estimation, rather than the similarity or rigid-body alignments conventionally used for monocular or loosely-constrained systems. On a visible-light baseline the estimator recovers the reference trajectory to sub-metre accuracy, confirming that the implementation is correct. On thermal data, performance is strongly regime-dependent rather than uniformly degraded: an indoor sequence with ample parallax is tracked accurately but only briefly before the estimate diverges; a vehicle-mounted sequence with a distant, low-parallax scene is tracked for its full duration only with the more robust KLT front-end; and a slow-moving aerial sequence with near-zero parallax cannot be tracked at all, regardless of preprocessing or front-end choice. Three recurring failure modes are identified and tied to measurable evidence: triangulation starvation under low parallax, a front-end-dependent variant of the same starvation under marginal parallax, and brightness-constancy violation introduced by per-frame intensity normalisation—underlain throughout by weak scale observability under low platform excitation. A separate inertial-only test shows that IMU noise quality is not the dominant error source on most platforms studied, with one clear exception where an inadequate sensor model compounds an already low-parallax scene. The overall finding is that monocular thermal-inertial odometry with a classical filter-based pipeline is viable only within a narrow band of favourable operating conditions—sufficient parallax and platform excitation—and is not, on its own, a reliable metric odometry source under unconstrained motion. Stereo thermal sensing or fusion with an additional metric sensor is identified as the most direct way to address the recurring scale-related failure.

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1 Introduction

1.1 Motivation

Autonomous robots and unmanned aerial vehicles (UAVs) increasingly operate in environments where the Global Navigation Satellite System (GNSS) is unavailable or unreliable: indoors, underground, beneath dense canopy, and in disaster zones. In such settings, the platform must estimate its own motion from onboard sensors alone. Visual-inertial odometry (VIO), which fuses a camera with an inertial measurement unit (IMU), has become one of the dominant solutions to this problem, offering metrically-scaled, low-drift state estimation at low cost, weight, and power [1, 2, 3, 4].

The robustness of conventional VIO is fundamentally tied to the quality of the visual signal. Standard cameras operate in the visible spectrum and therefore degrade or fail in exactly the conditions that motivate autonomy in the first place: darkness, smoke, dust, fog, and glare. Search-and-rescue, firefighting, subterranean exploration, and night-time operations all routinely present these conditions, in which a visible-light front-end loses the texture it depends on.

Long-wave infrared (LWIR) thermal cameras offer an alternative solution. By sensing emitted thermal radiation rather than reflected visible light, they perceive a scene in total darkness and see through smoke and light fog that would blind a conventional camera [5, 6]. This makes thermal-inertial odometry (TIO) a promising direction for GNSS-denied navigation in adverse environments.

Thermal imaging, however, is not a drop-in replacement for a visible camera. The modality introduces several characteristics that directly challenge the assumptions of classical feature-based VIO front-ends:

- **Low spatial resolution and weak texture.** Thermal sensors typically provide far fewer pixels than visible cameras, and thermal scenes are often smooth and low-contrast, yielding few well-localised corners or trackable gradients. Furthermore, thermal appearance depends primarily on heat distribution rather than geometric texture, meaning that visually distinctive structures may provide less discriminative information in the thermal domain [7, 8].
- **Radiometric instability.** Internal non-uniformity correction (NUC)—periodically triggered by an internal shutter, and referred to as flat-field correction (FFC) on some commercial cores—together with automatic gain control (AGC) and scene temperature changes, alter the apparent intensity of a fixed physical point between frames. The remapping applied when high-bit-depth data is rescaled for 8-bit feature front-ends, together with the intensity discontinuity at each shutter event, violates the brightness-constancy assumption underlying optical-flow trackers [8].
- **High-bit-depth raw output.** Many LWIR thermal cameras provide raw radiometric output at 14- or 16-bit precision. Conventional feature detectors and descriptors, however, are typically designed for 8-bit images, requiring a radiometric normalisation step whose choice may affect image consistency across frames [7, 6].
- **Sensor noise and image artefacts.** Thermal sensors inherently suffer from higher noise levels than visible-light cameras [7, 6]. Furthermore, the finite thermal time

constant of the microbolometer detector introduces motion blur during fast movement, and commercial thermal devices used on UAVs often apply lossy on-board compression. Together, these factors inject additional uncertainty into feature measurements, severely degrading tracking robustness.

These properties mean that a pipeline which is correct for visible-light VIO is not guaranteed to perform on thermal data, and that the practical limits and operating conditions of monocular TIO remain an active area of research.

1.2 Aim of the Thesis

The aim of this thesis is to investigate the feasibility of monocular feature-based thermal-inertial odometry and to analyse the factors that influence its performance. Rather than proposing a new estimation algorithm, the work pursues a systematic and reproducible evaluation of an established visual-inertial filtering framework on the thermal modality. A visible-light visual-inertial baseline is used as a reference against which thermal performance can be interpreted. It also helps distinguish modality-specific effects from the generic behaviour of the estimator.

1.3 Thesis Outline

The remainder of this thesis is organised as follows.

Chapter 2 reviews related work in visual-inertial odometry, thermal-inertial odometry, and trajectory evaluation, and identifies the gap this thesis addresses.

Chapter 3 builds on that gap to state the problem, the research questions, the contributions, and the scope of the work.

Chapter 4 presents the theoretical background: the IMU motion model, the MSCKF formulation, feature tracking, and the error metrics used.

Chapter 5 describes the implemented pipeline in detail, including the thermal preprocessing, the front-end, the MSCKF back-end, initialisation, and the evaluation framework.

Chapter 6 introduces the datasets used in this work and their sensor suites.

Chapter 7 presents and discusses the per-dataset results, followed by a cross-dataset analysis of the observed trends and the limitations of monocular thermal-inertial odometry.

Chapter 8 concludes and outlines directions for future work.

2 Related Work

This chapter reviews the research relevant to monocular thermal-inertial odometry (TIO) and positions the present work within it. Visual-inertial odometry methods can be organised along two largely independent axes. The first is the *estimation back-end*, which traditionally falls into two geometric categories: *filtering-based*—recursive Kalman-filter variants that maintain a single current estimate—or *optimisation-based*, which jointly refine many states over a sliding window of keyframes. Recently, a third back-end paradigm has emerged: *direct motion regression*, which replaces geometric estimation entirely with end-to-end neural networks.

The second axis is the *visual front-end*, that is, how image information enters the estimator: *feature-based* (indirect) methods track sparse keypoints and descriptors; *direct* methods minimise photometric error on raw pixel intensities; and *learning-based* methods predict visual representations or features. Table 2.1 places the systems reviewed here in this two-dimensional space.

The two axes are largely but not fully independent: end-to-end motion regression is inherently learning-based, so that combination occupies a single cell rather than spanning the row and column.

Table 2.1: Taxonomy of the visual-inertial odometry methods reviewed in this chapter, by estimation back-end and visual front-end. Khattak et al. and DeepTIO operate on thermal imagery; the remaining systems were designed for visible-spectrum cameras. This thesis studies the filtering / feature-based combination (MSCKF) applied to the thermal modality.

Back-end Strategy	Feature-based	Direct	Learning-based
Filtering	MSCKF (This thesis)	ROVIO	—
Optimisation	VINS-Mono, PL-VIO, ORB-SLAM3	Khattak et al.	—
Direct Motion Regression	—	—	DeepTIO

The chapter proceeds from the methods most distant from the present work to those closest to it: optimisation-based estimators first (Section 2.1), then filtering-based estimators including the MSCKF used in this thesis (Section 2.2), then thermal-inertial odometry (Section 2.3). Section 2.4 summarises the trajectory-evaluation conventions adopted throughout, and Section 2.5 states the gap this work addresses.

2.1 Optimisation-based Visual-Inertial Odometry

Optimisation-based estimators formulate state estimation as a non-linear least-squares problem over a sliding window of recent states, jointly refining camera poses, velocities, biases, and feature parameters. This typically yields higher accuracy than recursive filtering at the cost of greater computational load. The systems below all use a feature-based (indirect) front-end.

2.1.1 VINS-Mono

VINS-Mono [2] adopts a tightly coupled non-linear optimisation framework. Instead of maintaining a state estimate only for the current timestep, the system performs estimation

over a sliding window of recent keyframes (typically 10–15 frames). Within this window, visual feature observations and inertial measurements are jointly optimised, enabling more accurate state estimation compared to purely filtering-based approaches.

However, this optimisation-based formulation introduces increased computational complexity, as multiple states and feature observations must be optimised simultaneously. In addition, the back-end requires a robust initialisation procedure to estimate the metric scale, gravity direction, and initial feature depths. To achieve reliable initialisation, the system typically requires sufficient excitation motion so that the IMU measurements provide enough information for accurate parameter estimation.

Furthermore, the system integrates additional modules such as relocalisation and global pose-graph optimisation, which enable long-term consistency and loop closure. Due to these components, the framework extends beyond pure odometry and shares many characteristics with a full SLAM system. While this design significantly improves robustness and global consistency, it also increases computational requirements compared to lightweight odometry pipelines.

2.1.2 PL-VIO

While conventional point-based VIO systems often degrade in environments with limited visual texture, PL-VIO [9] addresses this limitation by incorporating line-segment features into the state estimation process. The system operates within a tightly-coupled optimisation-based visual-inertial framework in which both point features and structural line segments are jointly estimated within a sliding window.

This dual-feature representation improves robustness where point features alone are insufficient, such as long corridors or scenes dominated by planar structures. Such conditions often lead to feature starvation, where traditional point detectors fail to extract a sufficient number of reliable keypoints. In contrast, line segments capture extended geometric structures and typically remain detectable even in low-texture regions, providing additional spatial constraints for the optimisation.

Although PL-VIO was originally developed for conventional visual imagery, the use of structural line features can also be beneficial in thermal imaging, where distinct corner features are often sparse. In thermal images, temperature discontinuities along structural elements—such as door frames or wall boundaries—can produce stable edge structures suitable for line-based feature extraction. Consequently, incorporating line features can improve odometry robustness in environments where reliable point features are limited.

2.1.3 ORB-SLAM3

While visual-inertial odometry focuses primarily on local state estimation, full Simultaneous Localisation and Mapping (SLAM) frameworks extend this capability by constructing a persistent map of the environment and performing loop closure to correct accumulated drift. A representative example is ORB-SLAM3 [4], which integrates visual and inertial measurements within a multi-map SLAM architecture. The system maintains a global pose graph and performs loop-closure detection and global bundle adjustment to ensure long-term consistency of the estimated trajectory.

While such capabilities significantly improve global accuracy and enable large-scale mapping, they also introduce additional computational and memory requirements due to the need for maintaining and optimising a global map representation. On resource-constrained Micro Aerial Vehicles (MAVs), particularly those equipped with thermal cameras, the on-board processor is often already heavily utilised by front-end image processing operations such as radiometric normalisation and feature extraction. Under these conditions,

lightweight odometry pipelines are often preferred in practice to ensure reliable real-time performance.

2.2 Filtering-based Visual-Inertial Odometry

Filtering-based estimators maintain a single recursive estimate, propagating it with the IMU and correcting it with visual measurements. They are computationally lighter than optimisation-based methods and therefore attractive for resource-constrained platforms. This is the paradigm adopted by the present work. The two systems below illustrate the two front-end styles within this family: a direct method (ROVIO) and a feature-based method (the MSCKF).

2.2.1 ROVIO: Robust Visual-Inertial Odometry Using a Direct EKF

ROVIO [3] integrates IMU and camera measurements within a tightly-coupled iterated Extended Kalman Filter (IEKF) framework. Inertial measurements are used during the prediction step, while visual observations provide corrective updates to the filter state. This tightly-coupled fusion allows the system to resolve the metric scale ambiguity inherent to purely monocular visual odometry, and to maintain tracking under low-parallax motion such as pure rotation.

Unlike traditional feature-based pipelines that rely on descriptor matching (e.g., ORB or SIFT), ROVIO follows a patch-based direct approach. While it extracts discrete keypoints to initialise tracking, it avoids descriptor computation entirely; instead, it tracks multi-level image patches around these keypoints by directly minimising the photometric error within the filter innovation.

To maintain robustness during aggressive motion and large inter-frame displacements, the system uses IMU-based motion prediction as a prior for the expected patch locations, together with a coarse-to-fine tracking strategy based on image pyramids. If a patch becomes significantly distorted or tracking fails, it is discarded to reject outliers.

2.2.2 MSCKF: A Multi-State Constraint Kalman Filter

The Multi-State Constraint Kalman Filter (MSCKF), introduced by Mourikis and Roumeliotis [1], is a foundational *feature-based* filtering approach and the back-end adopted in this thesis. Like other tightly-coupled methods it propagates the state with inertial measurements and corrects it with visual observations; its defining characteristic is the way it incorporates visual information *without* including landmark positions in the filter state. In conventional EKF-SLAM formulations every observed 3D feature is added to the state vector, causing its dimension—and therefore the computational cost—to grow without bound as the map expands.

The MSCKF avoids this growth by maintaining a sliding window of recent camera poses in the state instead of the landmarks. When a feature has been tracked across several of these poses and is about to leave the field of view, all of its observations are processed together to form a single geometric constraint linking those poses. The feature is triangulated and then algebraically eliminated from the update through a null-space projection, so it never enters the state vector. As a result the computational cost remains linear in the number of features and is bounded by the size of the pose window, which makes the filter well suited to resource-constrained platforms.

Because the MSCKF operates on sparse keypoint tracks rather than photometric patches, it combines the efficiency of a filtering back-end with conventional feature front-ends such as Kanade-Lucas-Tomasi (KLT) optical flow or ORB descriptors. A known limitation of the original formulation is estimator inconsistency arising from spurious information gain

along the unobservable directions of the problem (global position and yaw); subsequent work mitigates this through observability-constrained or First-Estimates Jacobian (FEJ) techniques [10, 11]. The pipeline developed in this thesis adopts the MSCKF for precisely these reasons—bounded computational cost and direct compatibility with standard thermal feature front-ends—and incorporates FEJ anchoring to preserve consistency.

2.3 Thermal-Inertial Odometry

The methods above were designed for visible-spectrum cameras. Thermal imaging introduces modality-specific challenges—low contrast, radiometric instability, and periodic sensor calibration—that have motivated dedicated thermal-inertial odometry systems. In the existing literature, two contrasting strategies have predominantly emerged to tackle these: a direct, photometric formulation and an end-to-end learning approach. Notably, the classical feature-based filtering paradigm—despite its dominance in visible-light odometry—has remained largely underexplored in the thermal domain, presenting a gap that this thesis explicitly targets through a tailored preprocessing and feature-tracking pipeline.

2.3.1 Keyframe-based Direct Thermal-Inertial Odometry

To address the challenges of thermal imaging, a direct, keyframe-based thermal-inertial odometry system was introduced in [7]. It employs a keyframe-based non-linear optimisation framework to jointly estimate the camera trajectory and inertial states while minimising photometric residuals. Unlike conventional pipelines that convert the sensor’s native 14-bit thermal images into an 8-bit representation, the proposed method directly processes the raw 14-bit radiometric data.

By operating directly on the raw radiometric data, the framework preserves the intensity information necessary for robust photometric alignment. In addition, thermal cameras periodically perform Flat-Field Correction (FFC) to compensate for internal sensor noise and non-uniformities. During this calibration the camera temporarily freezes and becomes visually unavailable, causing short intervals of sensor blindness. To mitigate this, the system relies on tightly coupled inertial measurements, which provide motion priors that allow the estimator to maintain continuous state propagation during these interruptions.

2.3.2 DeepTIO

A fundamentally different strategy replaces hand-designed geometric estimation with learned models. DeepTIO [12] is a deep-learning thermal-inertial odometry system that regresses motion directly from thermal image sequences and inertial measurements using neural networks, rather than tracking explicit features and propagating a filter.

Its central idea is *visual hallucination*: a network is trained to predict the visual (RGB) feature representation that would correspond to a given thermal input, transferring knowledge learned in the data-rich visible domain into the data-poor thermal domain. This compensates for the weak texture and low contrast that limit conventional thermal feature extraction. The approach is robust to several modality-specific artefacts that are difficult to model explicitly, but, like other learning-based methods, it depends on large quantities of training data, offers limited interpretability, and its accuracy on environments and platforms unseen during training is not guaranteed.

DeepTIO represents the learning-based branch of thermal-inertial odometry, in contrast to the classical, geometric pipelines discussed above. The present work follows the classical route, prioritising transparency and the ability to analyse how each component of the pipeline behaves under varying conditions.

2.4 Trajectory Evaluation

Quantitative comparison of an estimated trajectory against a reference relies on two standard metrics introduced for SLAM and odometry benchmarking [13]. The Absolute Trajectory Error (ATE) measures global consistency by aligning the estimated trajectory to the reference and computing the root-mean-square positional difference over the whole run. The Relative Pose Error (RPE) instead measures local drift by comparing relative motions over fixed sub-intervals, and is therefore less sensitive to a single global misalignment.

A subtle but important aspect is the choice of alignment. For monocular visual odometry, absolute scale is unobservable, so trajectories must be aligned with a similarity transform ($\text{Sim}(3)$) that includes scale. In visual-inertial estimation, however, the accelerometer and gravity render both metric scale and the global roll and pitch observable. Applying a scale-correcting $\text{Sim}(3)$ alignment would conceal the scale error that the inertial channel is meant to resolve, and similarly, applying a full 6-DOF rigid-body transform ($\text{SE}(3)$) would improperly absorb roll and pitch drift into the alignment. Therefore, the mathematically consistent approach recommended for VIO is a 4-DOF alignment (3D position and yaw) [14]. Accordingly, this thesis reports ATE under 4-DOF alignment; RPE, being computed from relative motions, is reported without alignment.

2.5 Positioning of this Thesis

Despite substantial progress in visual-inertial odometry, the majority of mature systems—whether filtering-based (MSCKF, ROVIO) or optimisation-based (VINS-Mono, PL-VIO, ORB-SLAM3)—are designed for visible-spectrum cameras and assume sufficient scene texture and a stable photometric appearance. Thermal-inertial odometry has so far been approached mainly through direct, photometric formulations on raw radiometric data [7, 15] or through learning-based estimators [12], and frequently relies on a stereo baseline or a single, carefully tuned platform. As Table 2.1 makes explicit, the filtering / feature-based combination—the MSCKF—is well established on visible-spectrum data but has not been systematically characterised on thermal imagery, particularly across heterogeneous platforms and under imperfect preprocessing and calibration.

This is the gap the present thesis addresses. The specific problem statement, the research questions that follow from this gap, the contributions, and the scope of the study are formulated in Chapter 3.

3 Problem Formulation

3.1 Problem Statement

As established in Chapter 2, thermal-inertial odometry (TIO) has so far been addressed mainly through direct, photometric formulations [7] and learning-based estimators [12], often on a stereo baseline or a single, carefully tuned platform. The classical, feature-based filtering combination, the Multi-State Constraint Kalman Filter (MSCKF) [1], is well established on visible-spectrum data but has not been systematically characterised on thermal imagery. It therefore remains unclear how strongly the performance of such a system depends on the operating regime (platform velocity and excitation, scene depth and parallax, thermal contrast) and on practical implementation factors (the normalisation of raw high-bit-depth thermal frames and the parametrisation of the IMU noise model).

This thesis addresses that gap by implementing a monocular MSCKF-based TIO pipeline and evaluating it under a common, reproducible protocol across four datasets (ROVTIO, FIReStereo, STheReo, and EuRoC) that span visible and thermal modalities, indoor and outdoor scenes, and aerial and ground platforms. A fifth dataset, AerialTN, is partially integrated but excluded from the core evaluation—owing to an unresolved IMU axis-convention issue and to its thermal stream being delivered as pre-compressed 8-bit video, which precludes the common preprocessing—and its full evaluation is left to future work. The visible-light dataset (EuRoC) is included as a modality-independent sanity check that establishes the baseline correctness of the estimator, thereby isolating the effects specific to the thermal modality.

3.2 Research Questions

The study is guided by a single overarching question:

Under what operating conditions can a classical, feature-based monocular MSCKF deliver reliable metric state estimation from thermal imagery and a low-cost IMU, and what factors limit its performance?

This question is decomposed into three sub-questions that structure the evaluation and the conclusions:

- RQ1 (Feasibility and baseline.)** Can the same MSCKF estimator that is validated on visible-light visual-inertial data produce metric odometry from raw thermal imagery and a low-cost IMU, and how large is the accuracy gap between the two modalities?
- RQ2 (Operating regime.)** How does tracking reliability depend on platform motion (velocity and acceleration excitation) and scene structure (depth and parallax, thermal contrast), and which regimes cause the estimator to diverge?
- RQ3 (Preprocessing and IMU noise parametrisation.)** How sensitive is estimation accuracy to (a) the thermal preprocessing chain (undistortion, denoising, contrast enhancement, and 8-bit normalisation) that prepares 14/16-bit raw frames for the feature front-end, including the ordering of its operators, and (b) the quality of the IMU noise parametrisation provided with each dataset? What failure modes dominate when these are imperfect?

For RQ3(b), the IMU noise model comprises the accelerometer and gyroscope noise densities and their random-walk (bias instability) terms. Reliable values are normally obtained

by characterising the specific sensor, for example through Allan variance analysis. In the datasets used here, some provide such characterised parameters while others supply only datasheet values; this thesis does not re-estimate them, but evaluates the pipeline across both cases. The aim is to assess to what extent an inadequate IMU noise parametrisation acts as a source of trajectory drift that is independent of the thermal front-end, an effect whose severity is found to vary across datasets.

3.3 Contributions

The contribution of this thesis is not a new estimation algorithm but a systematic and rigorous characterisation of monocular thermal-inertial odometry, supported by a reproducible evaluation framework. Specifically:

1. **A modular monocular MSCKF thermal-inertial pipeline** that cleanly decouples three layers (image preprocessing, the feature-tracking front-end with Kanade–Lucas–Tomasi (KLT) optical flow and ORB descriptors, and the MSCKF back-end), so that each can be analysed and tuned independently for the thermal modality.
2. **A consistent multi-dataset evaluation protocol.** The same estimator is run on four datasets with a uniform metric suite (Absolute Trajectory Error and Relative Pose Error after 4-DOF (position and yaw) alignment, drift rate, and filter-consistency indicators). A visible-light dataset (EuRoC [16]) serves as a modality-independent correctness baseline.
3. **A justified evaluation methodology for monocular visual-inertial estimation.** Because metric scale, roll, and pitch are observable in visual-*inertial* estimation while global position and yaw are not, this thesis argues for and uses 4-DOF trajectory alignment (over position and yaw) rather than scale-correcting Sim(3) alignment, which would mask the scale error the estimator is intended to recover, or full SE(3) alignment, which would absorb genuine roll and pitch drift.
4. **A sensitivity and failure-mode analysis.** The study empirically relates tracking performance to platform velocity, scene depth and parallax, thermal contrast, and the quality of the IMU noise parametrisation supplied with each dataset. Furthermore, it identifies recurring failure modes including brightness-constancy violation induced by per-frame normalisation, weak scale observability under low-acceleration excitation, and triangulation starvation under low parallax.
5. **A practical conclusion for deployment.** The evidence indicates that monocular TIO is viable only within favourable operating regimes and is not, on its own, a reliable metric odometry source under unconstrained conditions, which motivates sensor fusion or a stereo thermal baseline for real-world use.

3.4 Scope and Delimitations

To keep the study focused, the following boundaries are set:

- **Monocular only.** Although several datasets provide a stereo thermal pair, only a single (left) camera is used, so that the metric-scale behaviour of the inertial channel can be studied in isolation. Stereo is discussed only as future work.
- **Filter-based estimation.** The back-end is fixed to the MSCKF; optimisation-based estimators are reviewed (Chapter 2) but not re-implemented.
- **Fixed geometric calibration.** Camera intrinsics, camera–IMU extrinsics, and time offset are taken as provided by each dataset and treated as fixed; estimating or

perturbing them is out of scope. The only sensor-model quantities considered are the IMU noise parameters, whose adequacy varies between the datasets used.

- **Existing recordings.** Evaluation uses already-recorded public datasets and a collaborator-provided drone recording rather than a self-conducted flight campaign.
- **Offline evaluation.** The pipeline is run offline on recorded data; real-time and on-board deployment are out of scope.
- **Consistency evaluation.** Filter consistency is monitored through the χ^2 acceptance rate and the time-averaged NIS; NEES-based tests and formal NIS-band or Monte-Carlo consistency analyses are out of scope [17], as they require ground-truth state covariance or repeated runs.

4 Background

This chapter establishes the theoretical background required to follow the implementation in Chapter 5. It first fixes the notation and coordinate frames (Section 4.1), then develops the inertial motion model, the Multi-State Constraint Kalman Filter together with its observability properties, the visual front-end, thermal image formation and preprocessing, and finally the trajectory-evaluation metrics. The treatment is restricted to established results; the specific design choices and parameter values of the implemented system are deferred to Chapter 5.

4.1 Notation and Coordinate Frames

Three right-handed coordinate frames are used throughout this thesis. The *world* frame $\{W\}$ is a fixed reference frame whose z -axis points upward, opposite to gravity. The gravity vector is written $\mathbf{g}^W = [0, 0, g]^\top$ with $g \approx 9.81 \text{ m/s}^2$, following the specific-force convention in which a stationary accelerometer reads $+g$ along its vertical axis: at rest the case housing pushes up on the proof mass, so the measured specific force balances gravity. For an ideal sensor with zero bias and noise, $\mathbf{R}_I^W \mathbf{a} = \mathbf{g}^W$, and the resulting coordinate acceleration $\dot{\mathbf{v}} = \mathbf{R}_I^W \mathbf{a} - \mathbf{g}^W$ vanishes. The *IMU* (body) frame $\{I\}$ is rigidly attached to the inertial sensor and is the frame in which the accelerometer and gyroscope measurements are expressed. The *camera* frame $\{C\}$ is centred at the optical centre with its z -axis along the optical axis. Since the camera is rigidly mounted to the IMU, the extrinsic transform $(\mathbf{R}_C^I, \mathbf{p}_C^I)$ is constant and is taken from calibration.

Scalars are written in italics (a), vectors in bold lowercase ($\mathbf{a}$) and matrices in bold upper-case ($\mathbf{A}$). A trailing superscript denotes the frame in which a quantity is expressed; for instance $\mathbf{p}^W$ is a point expressed in $\{W\}$. A rotation matrix $\mathbf{R}_B^A \in \text{SO}(3)$ maps a vector from frame $\{B\}$ to frame $\{A\}$, so that a point transforms as

$$\mathbf{p}^A = \mathbf{R}_B^A \mathbf{p}^B + \mathbf{p}_B^A, \quad (4.1)$$

where $\mathbf{p}_B^A$ is the position of the origin of $\{B\}$ expressed in $\{A\}$. Because $\mathbf{R}_B^A \in \text{SO}(3)$ is orthogonal, its inverse equals its transpose, $(\mathbf{R}_B^A)^{-1} = (\mathbf{R}_B^A)^\top = \mathbf{R}_A^B$. The skew-symmetric matrix of a vector $\boldsymbol{\omega} = [\omega_1, \omega_2, \omega_3]^\top$ is written $[\boldsymbol{\omega}]_\times$ and reproduces the cross product, $[\boldsymbol{\omega}]_\times \mathbf{v} = \boldsymbol{\omega} \times \mathbf{v}$:

$$[\boldsymbol{\omega}]_\times = \begin{bmatrix} 0 & -\omega_3 & \omega_2 \\ \omega_3 & 0 & -\omega_1 \\ -\omega_2 & \omega_1 & 0 \end{bmatrix}. \quad (4.2)$$

Orientation is represented by a unit quaternion under the Hamilton convention, $\mathbf{q} = [q_w, q_x, q_y, q_z]^\top$ with $\|\mathbf{q}\| = 1$; the implementation stores it in the $[q_x, q_y, q_z, q_w]$ ordering used by the SciPy library. Every unit quaternion corresponds to a rotation matrix $\mathbf{R}(\mathbf{q}) \in \text{SO}(3)$, and the two representations are used interchangeably. The exponential map relates a minimal rotation vector $\boldsymbol{\phi} \in \mathbb{R}^3$ to a rotation,

$$\text{Exp}(\boldsymbol{\phi}) = \exp([\boldsymbol{\phi}]_\times) \in \text{SO}(3), \quad (4.3)$$

with inverse $\text{Log}(\cdot)$. For a small rotation, the first-order approximation $\text{Exp}(\delta\boldsymbol{\theta}) \approx \mathbf{I} + [\delta\boldsymbol{\theta}]_\times$ is used when linearising the propagation and measurement Jacobians in Chapter 5; the error-state formulation and the rotation conventions adopted throughout follow [18].

Orientation is propagated by right-multiplication with a body-frame increment: for a body-frame angular velocity ω integrated over an interval Δt ,

$$\mathbf{R}_{I,k+1}^W = \mathbf{R}_{I,k}^W \text{Exp}(\omega \Delta t). \quad (4.4)$$

The estimator maintains a nominal state together with a minimal error state. For a quantity $\mathbf{x}$ the estimate is denoted $\hat{\mathbf{x}}$, and the error is defined additively, $\mathbf{x} = \hat{\mathbf{x}} + \tilde{\mathbf{x}}$, for every quantity except orientation, whose error is a body-frame rotation applied multiplicatively on the right,

$$\mathbf{R}_I^W = \hat{\mathbf{R}}_I^W \text{Exp}(\delta\theta), \quad (4.5)$$

so that the three-parameter $\delta\theta$ captures the attitude error without the redundancy of the four-parameter quaternion. The inertial error state stacks fifteen degrees of freedom,

$$\tilde{\mathbf{x}}_I = [\delta\theta^\top, \delta\mathbf{b}_g^\top, \delta\mathbf{v}^\top, \delta\mathbf{b}_a^\top, \delta\mathbf{p}^\top]^\top \in \mathbb{R}^{15}, \quad (4.6)$$

comprising the attitude error $\delta\theta$ and gyroscope-bias error $\delta\mathbf{b}_g$ (expressed in $\{I\}$), the velocity error $\delta\mathbf{v}$ and position error $\delta\mathbf{p}$ (expressed in $\{W\}$), and the accelerometer-bias error $\delta\mathbf{b}_a$ (expressed in $\{I\}$). This fixed ordering is used in the propagation and measurement Jacobians of Chapter 5.

4.2 Inertial Measurement Model

Between visual updates the MSCKF propagates its state by integrating the inertial measurements. This section states the IMU measurement model, the resulting state kinematics, and the linearised error-state propagation that carries the covariance forward; the implementation specifics (integration scheme and numerical noise values) are deferred to Chapter 5. The formulation follows the error-state convention of [18].

4.2.1 Sensor Measurement Model

A low-cost inertial measurement unit provides the body angular rate from its gyroscope and the specific force from its accelerometer, each corrupted by a slowly varying bias and additive white noise:

$$\omega_m = \omega + \mathbf{b}_g + \mathbf{n}_g, \quad (4.7)$$

$$\mathbf{a}_m = \mathbf{a} + \mathbf{b}_a + \mathbf{n}_a, \quad (4.8)$$

where ω is the true angular rate and $\mathbf{a}$ the true specific force, both expressed in $\{I\}$. The accelerometer senses specific force rather than coordinate acceleration and therefore cannot separate gravitational from inertial effects; at rest $\mathbf{a} = (\mathbf{R}_I^W)^\top \mathbf{g}^W$, which recovers the static condition of Section 4.1. The biases $\mathbf{b}_g$ and $\mathbf{b}_a$ are modelled as random walks driven by $\mathbf{n}_{bg}$ and $\mathbf{n}_{ba}$, while $\mathbf{n}_g$ and $\mathbf{n}_a$ are white noise. These four stochastic terms are characterised by the IMU noise parameters: the gyroscope and accelerometer noise densities σ_g, σ_a and the bias random-walk coefficients σ_{bg}, σ_{ba} . Their values are taken from each dataset (Chapter 6), and the sensitivity of the estimate to their adequacy is one focus of this study.

4.2.2 State Kinematics and Propagation

The equations below describe the evolution of the true state, in which the bias-corrected measurements $\omega = \omega_m - \mathbf{b}_g$ and $\mathbf{a} = \mathbf{a}_m - \mathbf{b}_a$ use the true biases (the zero-mean white noise is omitted from the deterministic update and enters only through the process noise).

They drive the continuous-time kinematics of the inertial state:

$$\dot{\mathbf{R}}_I^W = \mathbf{R}_I^W [\boldsymbol{\omega}]_{\times}, \quad (4.9)$$

$$\dot{\mathbf{v}} = \mathbf{R}_I^W \mathbf{a} - \mathbf{g}^W, \quad (4.10)$$

$$\dot{\mathbf{p}} = \mathbf{v}, \quad (4.11)$$

$$\dot{\mathbf{b}}_g = \mathbf{n}_{bg}, \quad \dot{\mathbf{b}}_a = \mathbf{n}_{ba}, \quad (4.12)$$

with $\mathbf{g}^W = [0, 0, g]^\top$ as fixed in Section 4.1. The orientation evolves on $\text{SO}(3)$ through the body angular rate (Eq. 4.4); the velocity through the specific force rotated into $\{W\}$ minus gravity; and the position through the velocity. Integrating over the inter-measurement interval Δt with a first-order (Euler) scheme gives the discrete propagation that constitutes the filter's prediction step:

$$\mathbf{R}_{k+1} = \mathbf{R}_k \text{Exp}(\boldsymbol{\omega} \Delta t), \quad (4.13)$$

$$\mathbf{v}_{k+1} = \mathbf{v}_k + (\mathbf{R}_k \mathbf{a} - \mathbf{g}^W) \Delta t, \quad (4.14)$$

$$\mathbf{p}_{k+1} = \mathbf{p}_k + \mathbf{v}_k \Delta t + \frac{1}{2} (\mathbf{R}_k \mathbf{a} - \mathbf{g}^W) \Delta t^2, \quad (4.15)$$

where $\mathbf{R}_k$ abbreviates $\mathbf{R}_{I,k}^W$. The exact integration scheme is specified in Chapter 5. In practice the true biases are unknown, so the filter evaluates these equations with the current bias estimates $\hat{\mathbf{b}}_g$ and $\hat{\mathbf{b}}_a$ in place of $\mathbf{b}_g$ and $\mathbf{b}_a$; the discrepancy between the true and estimated biases is exactly what the error state of Section 4.2.3 propagates.

4.2.3 Error-State Propagation

The filter does not track the uncertainty of the full nominal state directly, but the covariance $\mathbf{P}$ of the minimal error state $\tilde{\mathbf{x}}_I$ of Eq. 4.6. This covariance quantifies the filter's confidence in its estimate and grows during propagation as integrated IMU noise accumulates. Linearising the kinematics about the current estimate with the small-angle approximation $\text{Exp}(\delta\boldsymbol{\theta}) \approx \mathbf{I} + [\delta\boldsymbol{\theta}]_{\times}$ yields the continuous error dynamics:

$$\dot{\delta\boldsymbol{\theta}} = -[\boldsymbol{\omega}]_{\times} \delta\boldsymbol{\theta} - \delta\mathbf{b}_g - \mathbf{n}_g, \quad (4.16)$$

$$\dot{\delta\mathbf{v}} = -\mathbf{R}_I^W [\mathbf{a}]_{\times} \delta\boldsymbol{\theta} - \mathbf{R}_I^W \delta\mathbf{b}_a - \mathbf{R}_I^W \mathbf{n}_a, \quad (4.17)$$

$$\dot{\delta\mathbf{p}} = \delta\mathbf{v}, \quad (4.18)$$

$$\dot{\delta\mathbf{b}}_g = \mathbf{n}_{bg}, \quad \dot{\delta\mathbf{b}}_a = \mathbf{n}_{ba}. \quad (4.19)$$

The bias errors have no deterministic dynamics; they are driven purely by their random-walk noise and therefore contribute to the state only through the process noise. The coefficients of $\delta\boldsymbol{\theta}$, $\delta\mathbf{v}$, $\delta\mathbf{p}$, $\delta\mathbf{b}_g$ and $\delta\mathbf{b}_a$ in these relations populate the 15×15 continuous-time transition matrix $\mathbf{F}$, while the white-noise terms $(\mathbf{n}_g, \mathbf{n}_a, \mathbf{n}_{bg}, \mathbf{n}_{ba})$ enter through a noise-input matrix $\mathbf{G}$. Together with the IMU noise parameters of Section 4.2.1, these define the discrete process-noise covariance $\mathbf{Q}$, which captures how much uncertainty the integrated inertial noise injects over one interval Δt . Discretising the dynamics to first order gives the state-transition matrix $\Phi = \mathbf{I} + \mathbf{F} \Delta t$, which maps the error state across one propagation step, and the covariance propagates as

$$\mathbf{P}_{k+1} = \Phi \mathbf{P} \Phi^\top + \mathbf{Q}. \quad (4.20)$$

The explicit construction of $\mathbf{G}$ and the discretisation of $\mathbf{Q}$ are given in Appendix A (Section A.1). To preserve estimator consistency, the rotation-dependent blocks of $\mathbf{F}$ are evaluated at a fixed first-estimate value rather than the latest estimate, as motivated in Section 4.3.5 and detailed in Chapter 5.

4.3 The Multi-State Constraint Kalman Filter

The MSCKF [1] fuses the inertial propagation of Section 4.2 with visual measurements while keeping the computational cost bounded. Its defining idea is to augment the state with a sliding window of recent camera poses and to treat each tracked feature as a geometric constraint linking the poses from which it was observed, rather than adding the feature to the state. The landmark is triangulated and then algebraically removed, so the map never enters the filter and the cost stays linear in the number of features.

4.3.1 State Representation

The filter state comprises the current inertial state of Section 4.2 augmented with a sliding window of N past camera poses. Each camera pose contributes an orientation $\mathbf{R}_{C_i}^W$ and a position $\mathbf{p}_{C_i}^W$, with the corresponding 6-DOF error $[\delta\boldsymbol{\theta}_{C_i}^\top, \delta\mathbf{p}_{C_i}^\top]^\top$, so the full error state and its covariance live in

$$\tilde{\mathbf{x}} = [\tilde{\mathbf{x}}_I^\top, \tilde{\mathbf{x}}_{C_1}^\top, \dots, \tilde{\mathbf{x}}_{C_N}^\top]^\top \in \mathbb{R}^{15+6N}. \quad (4.21)$$

Crucially, landmark positions are *not* part of the state. In EKF-SLAM every observed feature is appended to the state vector, so its dimension—and hence the update cost—grows without bound as the map expands. By retaining a fixed-size window of poses instead, the MSCKF keeps the state dimension bounded by N while still exploiting the multi-view information each feature carries. When a new image is recorded, the current camera pose—obtained from the inertial state through the fixed extrinsics $(\mathbf{R}_C^I, \mathbf{p}_C^I)$ —is appended to the window (state augmentation), and poses are marginalised once the window is full, keeping N constant; the specific pruning strategy is described in Chapter 5.

4.3.2 State and Covariance Propagation

Between images only the inertial state evolves; the camera poses already in the window are fixed in the world frame and do not change under IMU motion. Propagation therefore applies the inertial transition Φ_I of Section 4.2.3 to the IMU block of the covariance, leaves the camera blocks unchanged, and carries the IMU–camera cross-covariance through Φ_I :

$$\begin{bmatrix} \mathbf{P}_{II} & \mathbf{P}_{IC} \\ \mathbf{P}_{IC}^\top & \mathbf{P}_{CC} \end{bmatrix} \rightarrow \begin{bmatrix} \Phi_I \mathbf{P}_{II} \Phi_I^\top + \mathbf{Q}_I & \Phi_I \mathbf{P}_{IC} \\ \mathbf{P}_{IC}^\top \Phi_I^\top & \mathbf{P}_{CC} \end{bmatrix}. \quad (4.22)$$

State augmentation forms the new camera pose as a function of the current inertial state and the extrinsics; its covariance and its cross-covariance with the existing state are obtained by propagating $\mathbf{P}$ through the augmentation Jacobian, which contains, in particular, the lever-arm coupling $-\mathbf{R}_I^W [\mathbf{p}_C^I]_\times$ between the new camera position and the IMU attitude error.

4.3.3 Visual Measurement Model and Feature Triangulation

A feature f tracked across $M \leq N$ window poses yields one measurement per view. Expressing the landmark $\mathbf{p}_f^W$ in camera i and applying the pinhole projection $\pi(\cdot)$ gives the normalised image observation

$$\mathbf{z}_i = \pi((\mathbf{R}_{C_i}^W)^\top (\mathbf{p}_f^W - \mathbf{p}_{C_i}^W)) + \mathbf{n}_i, \quad \pi([X, Y, Z]^\top) = \frac{1}{Z}[X, Y]^\top, \quad (4.23)$$

with $\mathbf{n}_i$ the image-measurement noise, where the camera pose $(\mathbf{R}_{C_i}^W, \mathbf{p}_{C_i}^W)$ is the augmented window pose obtained from the inertial state through the extrinsics. Because the landmark is not in the state, it is recovered from its own observations. Holding the current window poses fixed, $\mathbf{p}_f^W$ is triangulated by minimising the summed reprojection error over the M views,

$$\mathbf{p}_f^W = \arg \min_{\mathbf{p}} \sum_{i=1}^M \|\mathbf{z}_i - \pi((\mathbf{R}_{C_i}^W)^\top (\mathbf{p} - \mathbf{p}_{C_i}^W))\|^2, \quad (4.24)$$

solved by Gauss–Newton iteration. To keep the problem well conditioned for distant points, the optimisation is carried out in an inverse-depth parameterisation, in which the depth is represented by its reciprocal so that a feature at large range maps to a small, finite quantity rather than diverging. The conditioning of Eq. 4.24 depends on the parallax, that is the angular spread of the viewing rays across the M poses: with wide parallax the rays intersect at a well-defined point, whereas under near-pure rotation or very distant structure the rays are almost parallel and the depth is poorly constrained. A minimum-parallax threshold therefore gates which tracks are triangulated, and the loss of parallax after a manoeuvre is one of the failure modes examined in Chapter 7.

4.3.4 Null-Space Projection and the Measurement Update

Linearising the M observations of feature f about the current estimate and stacking them gives

$$\mathbf{r} = \mathbf{H}_x \tilde{\mathbf{x}} + \mathbf{H}_f \delta \mathbf{p}_f^W + \mathbf{n}, \quad (4.25)$$

where $\mathbf{H}_x$ and $\mathbf{H}_f \in \mathbb{R}^{2M \times 3}$ are the Jacobians with respect to the state and the landmark position. The landmark error $\delta \mathbf{p}_f^W$ is a nuisance term that must not enter the state. It is eliminated by projecting the residual onto the left null space of $\mathbf{H}_f$: letting the columns of $\mathbf{A} \in \mathbb{R}^{2M \times (2M-3)}$ span $\text{null}(\mathbf{H}_f^\top)$ so that $\mathbf{A}^\top \mathbf{H}_f = \mathbf{0}$, left-multiplication of Eq. 4.25 by $\mathbf{A}^\top$ removes the landmark and reduces the $2M$ residuals to $2M - 3$ constraints on the camera poses alone,

$$\mathbf{r}_o = \mathbf{A}^\top \mathbf{r} = \mathbf{H}_o \tilde{\mathbf{x}} + \mathbf{n}_o, \quad \mathbf{H}_o = \mathbf{A}^\top \mathbf{H}_x. \quad (4.26)$$

This is the *multi-state constraint*. It is applied as a standard EKF update—with Kalman gain $\mathbf{K} = \mathbf{P} \mathbf{H}_o^\top (\mathbf{H}_o \mathbf{P} \mathbf{H}_o^\top + \mathbf{R}_o)^{-1}$, state correction $\mathbf{K} \mathbf{r}_o$, and the corresponding covariance update—after a χ^2 (Mahalanobis) test gates out tracks whose residual is inconsistent with the predicted covariance. The gating test itself is a Mahalanobis check on the projected residual: with innovation covariance $\mathbf{S} = \mathbf{H}_o \mathbf{P} \mathbf{H}_o^\top + \mathbf{R}_o$, a track is accepted only if

$$\gamma = \mathbf{r}_o^\top \mathbf{S}^{-1} \mathbf{r}_o < \chi_{1-\alpha}^2(d), \quad (4.27)$$

where d is the dimension of $\mathbf{r}_o$ and $\chi_{1-\alpha}^2(d)$ is the threshold of the chi-square distribution at confidence level $1 - \alpha$. Residuals that are large relative to their predicted covariance, typically from mistracked features that survived the front-end rejection, exceed the threshold and are discarded before they can corrupt the state. The feature is then discarded; as it never enters the state, the per-update cost remains bounded by the window size.

4.3.5 Observability and First-Estimates Jacobians

Visual-inertial estimation has four unobservable degrees of freedom: the global position (3) and the rotation about gravity, that is the yaw (1). Roll and pitch are rendered observable by the gravity direction sensed through the accelerometer, and metric scale by the accelerometer itself, but no measurement fixes the global position or heading. A consistent estimator must therefore gain no information along these directions. A standard EKF, however, re-evaluates its Jacobians at the latest estimate at every step; because the linearisation point changes over time, the linearised system acquires spurious observability of this subspace, the covariance shrinks along it, and the filter becomes over-confident—typically manifesting as spurious yaw drift [10].

The First-Estimates Jacobian (FEJ) technique [10, 11] prevents this by evaluating the propagation and measurement Jacobians at the *first* estimate of each state—fixed once that state enters the filter—rather than at the latest estimate. Holding the linearisation point fixed preserves the correct unobservable subspace, so no spurious information is gained and consistency is maintained. The pipeline developed in this thesis adopts FEJ anchoring; its concrete implementation is described in Chapter 5.

4.4 Visual Front-End

The front-end converts the raw image stream into the sparse feature tracks consumed by the measurement model of Section 4.3.3: each track is a sequence of normalised image observations of a single scene point across several frames. Two front-end paradigms are compared in this thesis—Kanade–Lucas–Tomasi (KLT) optical-flow tracking and ORB descriptor matching—because their differing assumptions are stressed in different ways by thermal imagery.

4.4.1 Feature Detection and Tracking

Salient points are first detected as corners: image locations with a strong intensity gradient in two directions, and therefore good localisation. The Shi–Tomasi criterion [19] is a standard choice. The two front-ends then establish inter-frame correspondences differently.

The KLT tracker [20] follows each corner from one frame to the next by directly aligning a small intensity window. Under the brightness-constancy assumption—that the appearance of a physical point is unchanged between consecutive frames—it solves for the displacement that minimises the photometric (sum-of-squared-difference) error, iterating from coarse to fine over an image pyramid to accommodate larger motions. It carries no descriptor, so a track that is lost cannot be re-acquired; new corners are detected to replenish the set.

ORB [21] instead detects FAST corners and computes a rotation-aware binary descriptor, associating features between frames by Hamming distance. Descriptor matching can re-associate a feature after larger inter-frame motion or a brief loss, at the cost of relying on the repeatability of the binary descriptor.

The two make different demands on the imagery, which matters for the thermal modality (Section 4.5): KLT depends on brightness constancy, which the radiometric instability of thermal sensors violates, whereas ORB depends on descriptor repeatability, which the low contrast of thermal scenes undermines. Each surviving track contributes its sequence of normalised observations to the multi-state constraint of Section 4.3.3.

4.4.2 Outlier Rejection

Incorrect correspondences would corrupt the filter, so a rejection chain precedes the update. For KLT, a forward–backward consistency check tracks each point to the next frame and back again; a point whose round-trip does not return to its origin within a tolerance is discarded, which catches asymmetric tracking failures. For descriptor matching, Lowe’s ratio test retains only matches whose best candidate is sufficiently closer than the second-best. A geometric test then verifies the surviving correspondences against a two-view model: a fundamental matrix is fitted with Random Sample Consensus (RANSAC) [22], and correspondences inconsistent with the recovered epipolar geometry are rejected as outliers, together with a gate on the maximum admissible pixel displacement. The cleaned tracks are passed to the back-end, where the χ^2 test of Section 4.3.4 provides a final statistical gate.

4.5 Thermal Image Formation and Preprocessing

Long-wave infrared (LWIR) thermal cameras image emitted thermal radiation (roughly 8–14 μm) rather than reflected visible light, so they perceive a scene in darkness and through smoke or light fog [7, 6]. A microbolometer sensor produces a radiometric frame at 14- or 16-bit depth, reflecting the wide dynamic range of scene temperature, which must be reduced to 8-bit before conventional detectors and descriptors can be applied.

Several characteristics of the modality challenge the brightness-constancy and repeatability assumptions of the front-end. Automatic gain control (AGC) rescales the displayed intensity of each frame according to the current scene-temperature range, so the apparent intensity of a fixed physical point changes between frames, directly violating brightness constancy [8]. Periodic non-uniformity correction (NUC), which on many commercial cores is performed by briefly closing an internal shutter and is then also referred to as flat-field correction (FFC), recalibrates the per-pixel response [5]; during this interval the image freezes and the camera is momentarily blind. The sensors are also of low spatial resolution, and thermal scenes are often smooth and low-contrast, yielding few well-localised corners.

To prepare the raw frames, a sequence of established operators is applied: geometric undistortion (a plumb-bob or fisheye remap), Gaussian denoising to suppress sensor speckle, Contrast-Limited Adaptive Histogram Equalisation (CLAHE) [23] for local contrast enhancement with a clip limit that bounds noise amplification, and a final normalisation of the high-bit-depth frame to 8-bit by fixed or per-frame percentile bounds. The choice and *ordering* of these operators is not unique; their effect on tracking, in particular the relative placement of contrast enhancement and normalisation, is examined empirically in this thesis (Chapter 7).

Because contrast enhancement is central to preparing low-contrast thermal frames for feature extraction, the CLAHE operator is described in more detail. Ordinary histogram equalisation spreads the intensity distribution of an image to increase global contrast: with $h(i)$ the histogram of an L -level image of n pixels, the normalised cumulative distribution

$$T(i) = \frac{L-1}{n} \sum_{j=0}^i h(j) \quad (4.28)$$

is used as an intensity mapping, so that a flat region of the histogram is stretched and faint gradients become steeper. Applying this transform globally, however, over-amplifies noise in near-uniform regions and washes out local structure. *Adaptive* histogram equalisation instead partitions the image into small tiles and computes a separate mapping $T(i)$ per tile, enhancing local rather than global contrast, which suits thermal scenes whose informative gradients are weak and spatially localised. The *contrast-limited* variant additionally clips each tile histogram at a threshold (the clip limit) before forming the cumulative sum, redistributing the clipped mass uniformly; this bounds the slope of $T(i)$ and therefore the amount of noise amplification. The net effect is to sharpen the weak edges on which the feature front-end depends, at the cost of a tunable trade-off between contrast gain and noise, whose placement in the preprocessing chain is examined in Chapter 7.

4.6 Trajectory-Evaluation Metrics

Estimated trajectories are assessed against a reference using two complementary metrics [13]. The Absolute Trajectory Error (ATE) measures global consistency: after aligning the estimate to the reference, it reports the root-mean-square position difference over the whole run. The Relative Pose Error (RPE) measures local drift by comparing relative motions over fixed sub-intervals; being a function of relative poses, it is invariant to any global alignment.

The choice of alignment must match the unobservable degrees of freedom of the sensor configuration [14]. For monocular visual odometry the absolute scale and the global rigid pose are unobservable (7 DOF: 3 position, 3 orientation, and 1 scale), so a similarity

transform $\text{Sim}(3)$ is appropriate. In visual-*inertial* estimation, however, the accelerometer renders metric scale observable and gravity renders roll and pitch observable, leaving only the global position and the yaw—four degrees of freedom—unobservable. The aligning transform must therefore be restricted to exactly these four: a 4-DOF alignment over position and yaw. A full $\text{SE}(3)$ alignment would additionally rotate the trajectory in roll and pitch and so absorb genuine attitude drift, whereas a $\text{Sim}(3)$ alignment would rescale it and conceal the very scale error the inertial channel is meant to resolve; both would flatter the estimate. Accordingly, this thesis reports ATE under 4-DOF alignment and RPE—being alignment-invariant—directly, computed with the `evo` library [24]. These are complemented by the drift rate (ATE as a fraction of the path length) and a filter-consistency indicator based on the normalised innovation squared (NIS). The NIS reuses the Mahalanobis quantity of Eq. 4.27, $\gamma = \mathbf{r}_o^\top \mathbf{S}^{-1} \mathbf{r}_o$, whose expectation under a consistent filter equals the residual dimension d [17]. A time-averaged NIS substantially above d signals an over-confident filter (covariance too small, valid updates rejected), while a value well below d signals an over-conservative one; the indicator therefore reveals whether the IMU noise parametrisation of Section 4.2.1 matches the data.

5 Method

This chapter describes the implemented monocular thermal-inertial odometry (TIO) pipeline concretely, from thermal preprocessing through the feature front-end and the MSCKF back-end, to initialisation and the evaluation framework.

5.1 System Overview

The implemented system is a monocular thermal-inertial odometry pipeline that consumes a stream of images and inertial measurements and produces a metric trajectory estimate. It is organised into three deliberately decoupled layers. The *preprocessing* layer turns each raw high-bit-depth frame into an 8-bit image suitable for the front-end. The *front-end* detects and tracks features across frames, producing feature tracks—sequences of image observations of the same scene point—after an outlier-rejection stage. The *back-end* is the MSCKF of Section 4.3, which propagates the state with the inertial measurements and updates it with the feature tracks. Images and inertial samples are associated by timestamp, with the filter propagated over the inertial samples that fall between consecutive frames. The full pipeline is shown in Figures 5.1 and 5.2 in Section 5.2, which differ only in the ordering of the two preprocessing operators examined in that section.

The three layers communicate through narrow interfaces—an 8-bit image, a list of feature tracks, and the filter state—so that each can be modified or replaced without touching the others. This modularity is deliberate: it lets the front-end be switched between KLT and ORB, the preprocessing operators and their ordering be varied, and the back-end parameters be tuned, all independently, which is precisely what the sensitivity study of Chapter 7 exploits. The pipeline is implemented in Python, using OpenCV for the image operations and feature front-ends and NumPy/SciPy for the filter linear algebra. The implementation is publicly available at <https://github.com/akinmgumus/Monocular-Thermal-Inertial-Odometry>. Only a single (left) camera is used, and the system is run offline on recorded data; a deployed instance would output the same raw metric trajectory online. The remaining sections describe each layer in turn: thermal preprocessing (Section 5.2), the feature front-end (Section 5.3), the MSCKF back-end (Section 5.4), initialisation (Section 5.5), zero-velocity updates (Section 5.6), the filter and front-end parameters (Section 5.7), and the evaluation framework (Section 5.8).

5.2 Thermal Preprocessing

Each raw thermal frame passes through a fixed chain of operators before reaching the front-end: undistortion, Gaussian denoising, and—in one of two orderings—contrast enhancement and 8-bit normalisation. Both orderings were implemented; the full pipeline is shown for each in Figures 5.1 and 5.2.

Undistortion. Raw thermal frames exhibit lens distortion whose magnitude depends on the specific camera and lens, which would otherwise make the apparent motion of a feature inconsistent from one frame to the next. Each frame is undistorted with a precomputed remap look-up table built from the calibrated intrinsics $\mathbf{K}$ and distortion coefficients $\mathbf{D}$ (a plumb-bob or fisheye model, depending on the dataset), so that feature motion reflects camera motion alone. The remap keeps $\mathbf{K}$ unchanged, so tracked pixels can be mapped directly to normalised bearings with $\mathbf{K}^{-1}$ in the measurement model of Section 4.3.3.

Denoising. Thermal sensors exhibit per-pixel speckle noise. A Gaussian blur (kernel size 3×3) is applied before contrast enhancement, so that the enhancement step does

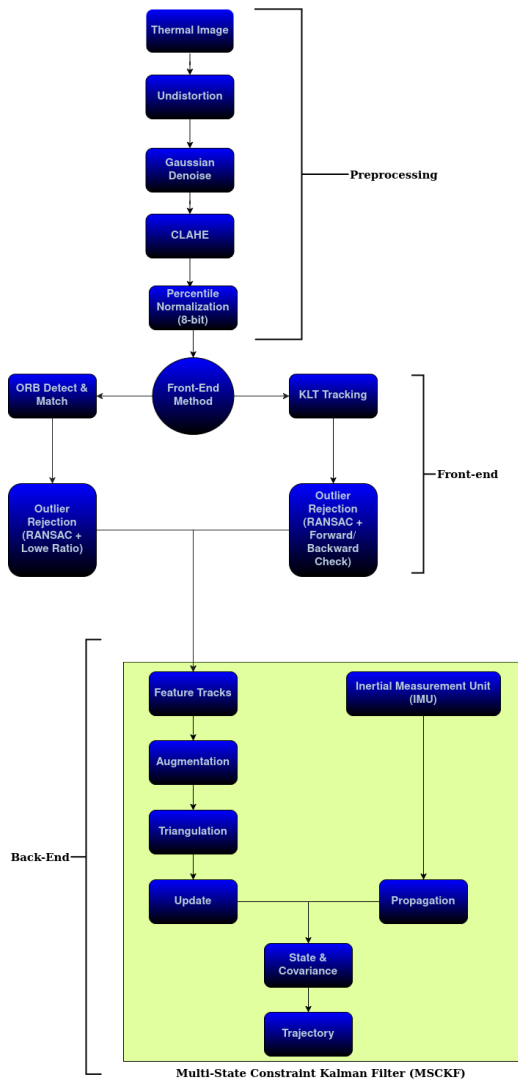


Figure 5.1: MSCKF pipeline—CLAHE before normalisation.

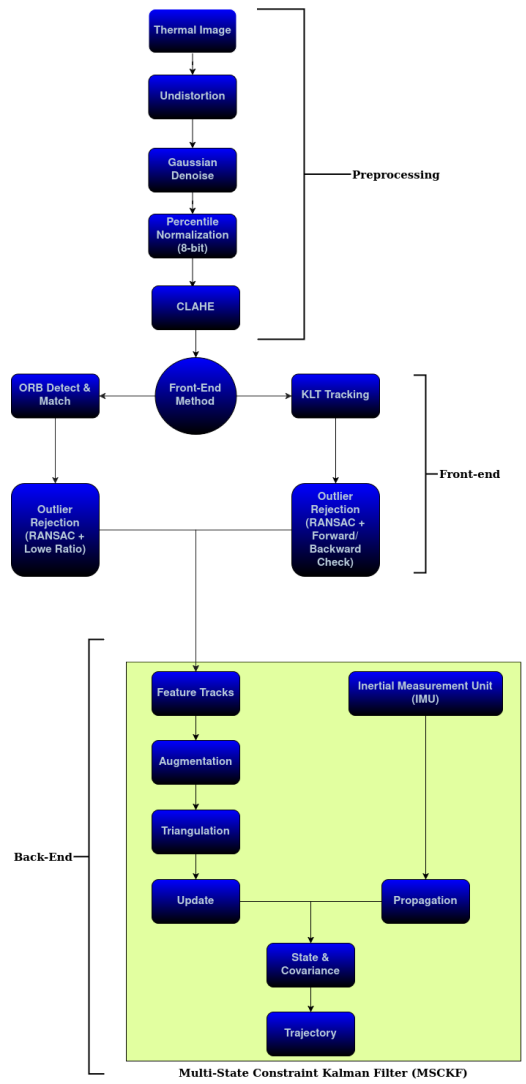


Figure 5.2: MSCKF pipeline—CLAHE after normalisation.

not amplify this noise into spurious gradients that would otherwise be picked up as false corners by the front-end.

Contrast enhancement and normalisation. The sensor’s native 14- or 16-bit radiometric output must be mapped to the 8-bit range expected by the front-end, and its native contrast is low. Contrast-Limited Adaptive Histogram Equalisation (CLAHE) is applied with a clip limit of 3.0 on an 8×8 tile grid, and the 8-bit mapping is obtained by a per-frame percentile normalisation (clipping to the 2nd and 98th percentile of the frame). These two operators do not commute: CLAHE’s clip limit is defined relative to the histogram it operates on, so applying it before or after the bit-depth reduction changes its effect. Both orderings were therefore implemented, as separate, otherwise identical preprocessing paths, and compared empirically; the comparison and the ordering adopted for the final results are reported in Section 7.3. Figure 5.3 illustrates the two intensity-mapping stages of this chain—normalisation and CLAHE—on a representative frame; the undistortion and Gaussian denoising applied beforehand are not perceptible at this scale and are omitted.

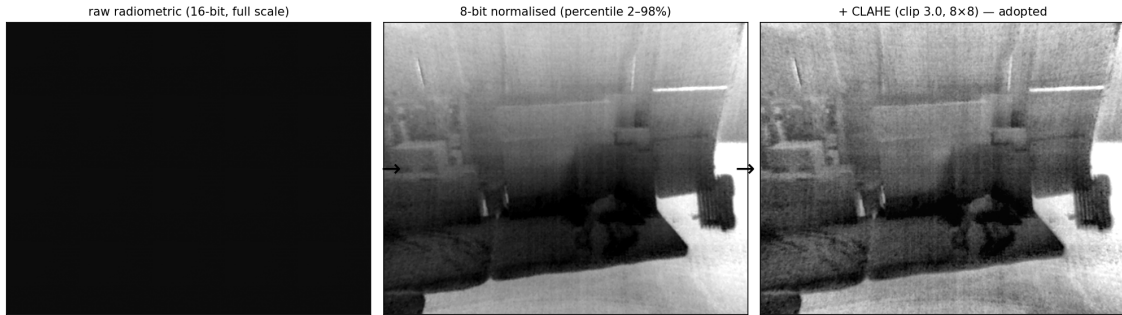


Figure 5.3: The intensity-mapping stages of the adopted Norm→CLAHE preprocessing chain on one ROVTIO frame; the undistortion and 3×3 Gaussian denoising applied first are imperceptible on the raw frame and are omitted here. The raw radiometric values occupy only a thin slice of the 16-bit range (here $\approx 3150\text{--}3327$ of $0\text{--}65535$), so shown on the full scale the frame reads as nearly flat and dark (left)—which is precisely why an 8-bit mapping is needed. Per-frame percentile (2–98%) normalisation stretches that slice across the 8-bit range (centre), recovering the global dynamic range, and CLAHE (clip 3.0, 8×8) then adds the local contrast and edge detail the front-end relies on (right).

5.3 Feature Front-End

Two front-end variants are implemented and compared in this thesis: Kanade–Lucas–Tomasi (KLT) optical-flow tracking and ORB detection and descriptor matching. Both produce the same output—a set of feature tracks, each a sequence of pixel observations of one scene point across consecutive frames—so either can be fed to the back-end of Section 5.4 without modification. Both variants share a geometric outlier-rejection stage—a maximum-displacement gate and a RANSAC fundamental-matrix test—but each additionally applies a check specific to its matching mechanism: a forward–backward consistency check for KLT and Lowe’s ratio test for ORB, as detailed in the subsections below. The gates specific to triangulation—parallax, depth, and the χ^2 consistency test—are deferred to the back-end, where they act once a track’s observations are used in an update.

The fundamental matrix is used only for geometric verification of correspondences, not for

pose recovery, which comes from the IMU–MSCKF fusion; a calibration-free fundamental-matrix fit is therefore sufficient and no essential-matrix decomposition is performed. The eight-point fundamental-matrix estimate degenerates under low parallax, pure rotation, or planar scenes, where a near-static baseline makes the epipolar geometry ill-defined and the test can admit spurious inliers; in this regime the parallax and χ^2 gates of the back-end (Section 5.4) act as the primary safeguard.

5.3.1 Kanade–Lucas–Tomasi Tracking

KLT tracking follows a feature by directly matching local pixel intensities rather than a descriptor. Features are first located as Shi–Tomasi corners [19]: image points with a strong intensity gradient in two directions, and therefore well localised. Each corner is then followed from one frame to the next with pyramidal Lucas–Kanade optical flow [20], which estimates, for a small window centred on the point, the sub-pixel displacement that best aligns the window’s intensity pattern between the two frames; a coarse-to-fine image pyramid allows this to accommodate larger inter-frame motions than a single-resolution search.

Because thermal imagery often has low geometric texture, few strong corners may be available; where they are, KLT is well suited to tracking them since it matches intensities directly and does not depend on a repeatable descriptor, unlike ORB’s binary descriptor, which is more sensitive to the modality’s contrast and noise. Its corresponding weakness is that it relies on brightness constancy—the assumption that a physical point’s apparent intensity is unchanged between frames—which the radiometric instability of thermal sensors (Section 4.5) can violate.

Two checks reject unreliable tracks before they reach the back-end. A forward–backward consistency check tracks each point forward from frame $n - 1$ to n and then backward from n to $n - 1$; if the round trip does not return close to the original pixel, the correspondence is discarded. A maximum-displacement gate then rejects any correspondence whose pixel motion exceeds a fixed bound. The surviving correspondences across the whole frame are finally fitted jointly to a fundamental-matrix model with RANSAC [22]; points inconsistent with this common two-view geometry are rejected as outliers, rather than being judged individually. Because KLT carries no descriptor, a track that is lost cannot be re-acquired: whenever the number of active tracks falls below a set threshold, new Shi–Tomasi corners are detected in the regions not already covered by a track, replenishing the set. The surviving tracks are passed to the back-end.

5.3.2 ORB Detection and Matching

ORB tracking instead re-identifies features by descriptor matching rather than following pixel intensities. In each frame, FAST corners are detected and a 256-bit binary BRIEF descriptor [21] is computed for each; detection is performed on a grid of cells covering the image, retaining only the strongest corners per cell, so that features are not concentrated in the highest-contrast regions of the frame at the expense of coverage elsewhere.

Correspondences with the previous frame’s tracked descriptors are established by Hamming-distance matching. For robustness, matching is run in both directions—from the previous frame’s descriptors to the current frame’s and back—and a correspondence is kept only if it is each other’s nearest neighbour in both directions (mutual nearest-neighbour consistency) and if its distance is sufficiently smaller than that of the second-nearest candidate (Lowe’s ratio test), which rejects ambiguous matches in repetitive or low-texture regions. As with KLT, the surviving correspondences are then passed through the same maximum-displacement gate and a joint RANSAC fundamental-matrix test before being accepted as tracks.

Because matching is descriptor-based rather than sequential, ORB can in principle re-associate a feature after a brief loss without re-detection. This comes at the cost of depending on the descriptor’s repeatability, which the low native contrast and higher noise of thermal imagery (Section 4.5) can undermine, making correct matches harder to distinguish from incorrect ones by Hamming distance alone.

5.4 MSCKF Back-End

The back-end implements the MSCKF of Section 4.3 with the concrete choices described here. The inertial state is propagated between images by the prediction of Section 4.2.2, and the sliding window is bounded to a fixed number of camera poses $N_{\max}$.

State augmentation. When an image is received, the current camera pose is computed from the inertial state and the fixed extrinsics, $\mathbf{R}_C^W = \mathbf{R}_I^W \mathbf{R}_C^I$ and $\mathbf{p}_C^W = \mathbf{p}_I^W + \mathbf{R}_I^W \mathbf{p}_C^I$, and appended to the state. This is a deterministic cloning of the current pose rather than a prediction or an update: it enlarges the dimension of the state and covariance without injecting any process noise, so the estimate’s uncertainty is not increased, only extended to the new pose. The new block is formed through the augmentation Jacobian, whose key coupling term is the lever-arm block $-\mathbf{R}_I^W [\mathbf{p}_C^I]_{\times}$, which expresses how the new camera position depends on the IMU attitude error.

Measurement update. A feature is used to update the filter only once it can no longer be tracked—when its track terminates or its earliest observation is about to leave the sliding window—so that its full multi-view history is available. Algorithm 1 summarises the procedure. Each mature track is first triangulated from the observations still in the window—a linear direct-linear-transform (DLT) initialisation, solved by the singular-value decomposition, refined by an inverse-depth Gauss–Newton minimisation of the reprojection error; tracks with insufficient parallax, or with a triangulated depth outside a plausible range, are rejected because they yield ill-conditioned constraints. The measurement Jacobians are then formed and the landmark is removed by the null-space projection of Section 4.3.4, giving the pose-only residual of Eq. 4.25. Each track is gated by a χ^2 (Mahalanobis) test against its predicted covariance before its constraint is stacked, and to bound the update cost at most a fixed number of the longest tracks are used per update. The stacked constraints are applied as a single EKF update.

In Algorithm 1, $\theta_{\min}$ is the minimum parallax required for a stable triangulation, $[d_{\min}, d_{\max}]$ the admissible landmark-depth range, and γ_{χ^2} the χ^2 threshold at confidence α for the residual’s degrees of freedom (Section 4.3.4). $\mathbf{R}$ is the image-measurement noise covariance, $\mathbf{S} = \mathbf{H}_o \mathbf{P} \mathbf{H}_o^T + \mathbf{R}$ the per-track innovation covariance, and $N_{\max}$ the window size. The covariance update uses the Joseph form, which preserves symmetry and positive-definiteness better than the shorter $(\mathbf{I} - \mathbf{K}\mathbf{H})\mathbf{P}$ expression. The operator $\boxplus$ injects the error-state correction into the nominal state: additively for position, velocity and the biases, and multiplicatively (Eq. 4.5) for the orientations. The numerical values of all these parameters are collected in Section 5.7.

Window management. After each update the oldest camera poses are marginalised so that the window never exceeds $N_{\max}$ (the PruneWindow step of Algorithm 1): their rows and columns are removed from the state and covariance, and the tracker is informed which poses have been dropped so that observations referring to them are no longer used.

First-estimate anchoring. To preserve the observability consistency discussed in Section 4.3.5, the rotation-dependent blocks of the propagation and measurement Jacobians are evaluated at first-estimate values rather than the latest estimate: the IMU orientation

Algorithm 1 MSCKF measurement update (one image).

Require: mature tracks $\mathcal{T}$, state $\hat{\mathbf{x}}$, covariance $\mathbf{P}$

```
1:  $\mathbf{H} \leftarrow []$ ,  $\mathbf{r} \leftarrow []$  ▷ stacked constraints
2: for each track  $f \in \mathcal{T}$  (longest first, up to a fixed limit) do
3:    $\mathcal{V} \leftarrow$  observations of  $f$  whose camera pose is still in the window
4:   if  $|\mathcal{V}| < 2$  or  $\text{parallax}(\mathcal{V}) < \theta_{\min}$  then
5:     reject  $f$ ; continue
6:   end if
7:    $\mathbf{p}_f \leftarrow \text{Triangulate}(\mathcal{V})$  ▷ inverse-depth Gauss–Newton
8:   if  $\text{depth}(\mathbf{p}_f) \notin [d_{\min}, d_{\max}]$  then
9:     reject  $f$ ; continue
10:  end if
11:  form  $(\mathbf{H}_x, \mathbf{H}_f, \mathbf{r}_f)$ : Jacobians at first-estimate poses, residual at current estimate
12:   $(\mathbf{H}_o, \mathbf{r}_o) \leftarrow \text{NullSpaceProject}(\mathbf{H}_x, \mathbf{H}_f, \mathbf{r}_f)$ 
13:  if  $\mathbf{r}_o^\top \mathbf{S}^{-1} \mathbf{r}_o > \gamma_{\chi^2}$  then
14:    reject  $f$ ; continue ▷  $\chi^2$  gate,  $\mathbf{S} = \mathbf{H}_o \mathbf{P} \mathbf{H}_o^\top + \mathbf{R}$ 
15:  end if
16:  stack  $(\mathbf{H}_o, \mathbf{r}_o)$  into  $(\mathbf{H}, \mathbf{r})$ 
17: end for
18: if  $\mathbf{H}$  is empty then return
19: end if
20:  $\mathbf{K} \leftarrow \mathbf{P} \mathbf{H}^\top (\mathbf{H} \mathbf{P} \mathbf{H}^\top + \mathbf{R})^{-1}$  ▷ Kalman gain
21:  $\hat{\mathbf{x}} \leftarrow \hat{\mathbf{x}} \boxplus \mathbf{K} \mathbf{r}$  ▷ inject correction (multiplicative on rotation)
22:  $\mathbf{P} \leftarrow (\mathbf{I} - \mathbf{K} \mathbf{H}) \mathbf{P} (\mathbf{I} - \mathbf{K} \mathbf{H})^\top + \mathbf{K} \mathbf{R} \mathbf{K}^\top$  ▷ Joseph form
23: PruneWindow to  $N_{\max}$  poses
```

entering the transition matrix is held at a fixed anchor, and each camera pose stores the estimate it had at augmentation, against which its measurement Jacobians are later linearised, while the residual itself uses the current estimate. In preliminary testing, chaining the camera anchors to the inertial anchor did not improve accuracy and was not adopted; each camera pose keeps its own independent first estimate.

5.5 Initialisation

The filter is initialised from a short static segment at the start of each sequence, during which the platform is stationary. Because the accelerometer then measures only the reaction to gravity, the direction of gravity in the body frame is recovered from the mean specific force over the segment, which fixes the initial roll and pitch (Section 4.1). This assumes the initial accelerometer bias is small, so that attitude is recovered from the mean specific force first and the residual then seeds the bias; the two are not separately observable from a static segment. The yaw is not observable from a static IMU—it is one of the unobservable directions of Section 4.3.5—and is therefore seeded from the reference trajectory where one is available; any residual yaw offset is absorbed by the 4-DOF alignment used in evaluation (Section 4.6), so this seeding does not affect the reported error. The initial position is taken from the reference and the initial velocity is set to zero.

The biases are estimated from the same static segment. The gyroscope bias is taken as the mean angular rate, which should be zero for a stationary sensor, and the accelerometer bias as the residual $\mathbf{a}_{\text{mean}} - (\mathbf{R}_I^W)^\top \mathbf{g}^W$ that remains after accounting for gravity in the recovered attitude. The initial-state covariance is set to reflect the confidence in each of these quantities.

This gravity-aligned initialisation is applied uniformly but adapted per dataset: platforms with a clean static start use it directly, while the yaw and position seeds are drawn from whichever reference each dataset provides (motion-capture, LiDAR-inertial, or the platform’s own onboard estimate). The length of the static segment is a per-dataset parameter (Section 5.7).

5.6 Zero-Velocity Updates

A visual-inertial estimator accumulates drift while the platform is stationary: with little parallax the visual updates are weak, and the inertial channel integrates its noise and residual biases unchecked. A zero-velocity update (ZUPT) counteracts this by supplying a direct pseudo-measurement that the velocity is zero whenever the platform is detected to be at rest. Rest is declared over a short sliding window of inertial samples when the mean angular rate is below a threshold and the mean specific-force magnitude is within a tolerance of g ; when both hold, the measurement $\mathbf{v} = \mathbf{0}$ with a small covariance is applied, throttled so that it is issued at most once every few samples.

ZUPT is implemented and available in the pipeline, but disabled for every reported run: enabling it selectively per dataset would conflict with the single shared parameter set used throughout (Section 5.7), and its rest detector risks corrupting genuine slow motion on several of the platforms studied here, which lack an extended static phase beyond the initialisation segment. It is retained as a pipeline feature and left to future work.

5.7 Filter and Front-End Parameters

Table 5.1 collects the tunable parameters of the pipeline together with their role. With the sole exception of the IMU noise parameters, every value is a single configuration shared across all datasets—deliberately, so that differences in tracking performance are

attributable to the data rather than to per-dataset tuning (Section 7.9 returns to the cost of this choice). The IMU noise densities and bias random-walk coefficients are the one quantity that is genuinely dataset-specific, taken from each dataset provider or datasheet as detailed in Chapter 6. Most of the shared values were tuned empirically to balance competing effects—for example the CLAHE clip limit trades contrast enhancement against noise amplification, and the χ^2 confidence trades outlier rejection against the risk of discarding valid measurements—rather than optimised by an exhaustive search.

Table 5.1: Tunable parameters of the pipeline, shared across all datasets except the IMU noise block, which is dataset-specific (Chapter 6).

Parameter	Symbol	Description	Value
<i>MSCKF back-end</i>			
χ^2 confidence	α	per-track Mahalanobis gate confidence	0.99
GN iterations	—	inverse-depth triangulation iterations	4
Min. parallax	$\theta_{\min}$	minimum bearing spread for triangulation	0.1°
Depth bounds	$d_{\min}, d_{\max}$	admissible landmark depth	0.1–200 m
Pixel noise	σ_{px}	image-measurement standard deviation	2.5 px
Window size	$N_{\max}$	camera poses in the sliding window	20
Max. tracks/update	—	longest tracks used per update	50
<i>KLT front-end</i>			
Corners	—	target number of tracked corners	700
FB tolerance	—	forward–backward round-trip bound	1.5 px
LK window	—	optical-flow window size	30×30 px
Pyramid levels	—	coarse-to-fine levels	3
Quality level	—	Shi–Tomasi corner-quality ratio	0.01
Min. distance	—	minimum spacing between corners	8 px
RANSAC threshold	—	fundamental-matrix inlier bound	2.0 px
Max. displacement	—	maximum inter-frame motion	150 px
<i>ORB front-end</i>			
Keypoints	—	target ORB keypoints	1500
Detection grid	—	grid of detection cells	4×4
Lowe ratio	—	nearest / second-nearest bound	0.85
FAST threshold	—	corner-detection threshold	7
Edge threshold	—	detector border margin	19
RANSAC threshold	—	fundamental-matrix inlier bound	3.5 px
Max. displacement	—	maximum inter-frame motion	150 px
<i>IMU noise (provenance in Chapter 6)</i>			
Noise densities	σ_g, σ_a	gyroscope / accelerometer white noise	dataset-specific
Bias random walk	σ_{bg}, σ_{ba}	gyroscope / accelerometer bias drift	dataset-specific

5.8 Evaluation Framework

Each run produces an estimated trajectory that is compared against the dataset’s reference. Following Section 4.6, the Absolute Trajectory Error is computed after a 4-DOF alignment over position and yaw, which matches the unobservable directions of visual-inertial estimation and therefore leaves any genuine roll, pitch or scale error visible in the metric. This 4-DOF alignment is implemented for this work, as the `evo` library [24] provides only $\text{SE}(3)$ and $\text{Sim}(3)$ Umeyama alignment; the Relative Pose Error, being in-

variant to any global alignment, is computed directly with `evo`. RPE is evaluated on the translational component only, over consecutive sub-segments spaced by 1 m of travelled reference distance (rather than by a fixed time or frame interval) and summarised by their RMSE; because the sub-segment length is exactly 1 m, this RMSE is reported as a distance-normalised quantity in metres of error per metre travelled (m/m), which remains comparable across sequences recorded at different speeds. These are complemented by the drift rate—the ATE as a fraction of the reference path length.

Because every sequence begins with a brief transient—an initial settling period during which the estimate drifts before locking onto the reference motion—and eventually diverges, three quantities are of interest: the full-trajectory ATE over the whole run, the ATE computed only over the tracking window between the end of the initial transient and the onset of divergence, and a survival time, measured from the start of the run to the onset of divergence, that quantifies how long the estimate tracks the reference before breaking down. The window boundaries are identified visually from the trajectory and error plots for each run, rather than by an automatic threshold: the observed failure pattern—an initial drift, a period of accurate tracking, and an eventual, non-recovering divergence—proved difficult to capture reliably with a fixed rule across datasets. These boundaries are recorded per run and applied at evaluation time (Chapter 7). For FReStereo, no tracking window could be identified; the estimate never locks onto the reference, and only the full-trajectory ATE is reported.

As a filter-consistency indicator, the time-averaged normalised innovation squared (NIS) is reported: the per-track Mahalanobis quantity $\mathbf{r}_o^T \mathbf{S}^{-1} \mathbf{r}_o$ of the χ^2 gate, divided by its degrees of freedom and averaged over the accepted tracks of the run. A value near one indicates a consistent filter; a value well above one indicates an over-confident filter (covariance too small), and well below one an over-conservative one. Because this average is taken over the tracks that pass the χ^2 gate, rejected outliers are excluded and the value is biased slightly downward; it is read as a relative indicator across datasets rather than an absolute test. The fraction of tracks accepted by the χ^2 gate is also reported as a coarser proxy for the same effect.

Each run is summarised by two figures: a *trajectory* figure showing the estimated and reference x , y and z components over time together with the top-down x – y path, and a *diagnostics* figure showing the position error over time alongside the number of active and used feature tracks. Axis limits are set from the reference so that it stays visible even when the estimate diverges, and the manually identified settle and divergence instants are marked with vertical lines. The diagnostics figure additionally shades the intervals where the instantaneous error exceeds a fixed, per-dataset threshold (the “seg thr” line) and reports them alongside a fixed-threshold segmentation of the run into tracking and diverged spans; this is a coarser, purely error-magnitude-based view kept for diagnostic reference, distinct from the manually identified tracking window used for the reported metrics—because it reacts to absolute error rather than to whether the estimate is still following the reference motion, it can shade large portions of an otherwise well-tracked run whose absolute error is inflated by a scale offset (Section 7.5), and should not be read as the tracking window itself. The per-run metrics are appended to a common results file, from which the cross-dataset comparison of Chapter 7 is produced.

6 Datasets

This chapter introduces the datasets used for evaluation, their sensor suites, ground-truth sources, and the motion profiles that define the operating regimes tested.

6.1 Overview and Sensor Suites

The pipeline is evaluated on four datasets spanning visible and thermal imagery, indoor and outdoor scenes, and aerial and ground platforms; a fifth (AerialTN) is partially integrated but excluded from the core evaluation (Section 6.6). This spread is deliberate: it exposes the estimator to a wide range of the operating regimes—platform velocity, scene depth, and thermal contrast—whose influence is the subject of RQ2, while the visible-light dataset serves as a modality-independent correctness baseline. Table 6.1 summarises the sensor suites and motion profiles; the individual datasets follow in the subsequent sections.

Table 6.1: Datasets used for evaluation. AerialTN is only partially integrated (Section 6.6); the left camera is used throughout.

Dataset	Modality	Camera	IMU	Reference / platform
EuRoC (MH_03)	visible	Aptina MT9V034, 752 × 480, 20 Hz	ADIS16448, 200 Hz	mocap; MAV, indoor
FIReStereo (frick_1)	thermal	Boson+, 640 × 512, 16 Hz	Epson G365, 200 Hz	LiDAR-inertial; UAS, outdoor
ROVTIO (alt1)	thermal	Tau2, 640 × 512, 25 Hz	VN-100, 168 Hz	Vicon; drone, indoor
STheReo (valley_evening)	thermal	FLIR A65, 640 × 512, 14-bit	Xsens MTi-300, 200 Hz	local pose; vehicle, outdoor
AerialTN	thermal	Purplerver Mini640, 640 × 512	472 Hz	PX4 EKF2; drone

6.2 EuRoC MAV (Visible-Light Baseline)

EuRoC [16] is a visible-light micro-aerial-vehicle dataset recorded indoors with synchronised global-shutter stereo cameras (Aptina MT9V034, WVGA 752 × 480 monochrome, 20 Hz; only the left camera is used) and an ADIS16448 IMU at 200 Hz, with millimetre-accurate motion-capture ground truth. It contains none of the thermal-specific artefacts of Section 4.5, and is therefore used here as a modality-independent sanity check: successful metric estimation on EuRoC establishes that the estimator itself is correct, so that any degradation on the thermal datasets can be attributed to the modality rather than to an implementation fault (RQ1). The MH_03 sequence, with moderate dynamics, is used.

6.3 FIReStereo

FIReStereo [6] is an outdoor thermal dataset recorded from an uncrewed aerial system over forest and urban scenes in visually degraded conditions. It provides a stereo FLIR Boson+ thermal camera (640 × 512, 16-bit, ~ 16 Hz; only the left camera is used), an Epson G365 IMU at 200 Hz, and a LiDAR-inertial trajectory used as pseudo-ground-truth—an estimate rather than an external reference, and treated accordingly. The frick_1 sequence is used; the platform moves slowly at altitude, giving the low parallax that stresses monocular triangulation.

6.4 ROVTIO

ROVTIO [25] is an indoor thermal-inertial dataset recorded on a small aerial platform. It provides a FLIR Tau2 thermal camera (640×512 , 16-bit, ~ 25 Hz) with a fisheye (equidistant) lens, a VectorNav VN-100 IMU at ~ 168 Hz, and Vicon motion-capture ground truth at ~ 6 Hz. The alt1 sequence begins with the platform stationary, which enables the gravity-aligned static initialisation of Section 5.5, and the indoor scene provides ample parallax, making it a well-conditioned indoor test case for examining both the regime in which monocular thermal odometry tracks and the point at which it diverges.

6.5 STheReo

STheReo [26] is a vehicle-mounted outdoor dataset providing 14-bit stereo thermal imagery from a FLIR A65 (640×512 ; only the left camera is used), an Xsens MTi-300 IMU at 200 Hz, and a reference trajectory supplied as a local pose (position and Euler angles). The valley_evening sequence is characterised by fast, largely forward motion at low thermal contrast—a combination adverse for monocular estimation, since forward motion yields little parallax on features near the direction of travel. It is included as an explicit negative-result case that marks an operating-regime limit of monocular thermal odometry (RQ2).

6.6 AerialTN

AerialTN is a thermal dataset recorded on a drone by a collaborator, using a Purplriver Mini640 thermal camera, with an IMU at 472 Hz and the platform’s own PX4 EKF2 estimate as a reference. It is excluded from the core evaluation for two reasons. First, an unresolved discrepancy in the IMU axis convention reported by the flight stack could not be reconciled with the other sensor streams, which prevents a correct inertial–visual fusion. Second, and more fundamentally for this study, the thermal imagery is delivered not as raw high-bit-depth frames but as an 8-bit, H.264-compressed video stream; the preprocessing chain central to the evaluation—the normalisation of the 14/16-bit radiometric output and the contrast-enhancement ordering of Section 5.2—therefore cannot be applied on the same footing as on the other datasets, and the lossy compression further degrades feature consistency, so any result would not be comparable. The dataset is nonetheless documented here, as it was a suggested candidate, and its integration scaffolding is retained for future work should the axis convention be confirmed and a raw stream become available.

6.7 Ground-Truth and Calibration Provenance

The datasets differ in the trustworthiness of their references, which matters when interpreting the reported errors. EuRoC and ROVTIO provide external motion-capture ground truth (Leica/Vicon), the most reliable; FIReStereo provides a LiDAR-inertial estimate and STheReo a local pose, both themselves the output of an estimator and therefore treated as pseudo-ground-truth rather than an absolute reference; AerialTN’s PX4 EKF2 estimate is the least authoritative. Camera intrinsics and camera–IMU extrinsics are taken as provided by each dataset—a plumb-bob model for EuRoC and FIReStereo, an equidistant fisheye model for ROVTIO—and treated as fixed (Section 3.4).

The IMU noise parameters are the only sensor-model quantities varied in this study (RQ3), and their provenance differs by dataset. EuRoC provides IMU noise parameters characterised for its specific unit and is treated as the characterised-reference case; for the thermal datasets (FIReStereo, ROVTIO, STheReo) only datasheet figures for the specific inertial unit (Epson G365, VectorNav VN-100, Xsens MTi-300) are available and are

used directly. Their adequacy therefore varies across datasets—characterised for EuRoC, datasheet-only for the thermal datasets—and its effect on trajectory drift and filter consistency is examined in Chapter 7. Table 6.2 gives the resulting values, used directly as $\sigma_g, \sigma_a, \sigma_{bg}, \sigma_{ba}$ in the process-noise covariance $\mathbf{Q}$ of Appendix A (Section A.1).

Table 6.2: IMU noise parameters used directly as $\sigma_g, \sigma_a, \sigma_{bg}, \sigma_{ba}$ in the process-noise covariance $\mathbf{Q}$ (Appendix A, Section A.1), without inflation. EuRoC’s are characterised for its specific unit; the remainder are datasheet figures for the named IMU.

Dataset	IMU	σ_g	σ_a	σ_{bg}	σ_{ba}
		[rad/s/ $\sqrt{\text{Hz}}$]	[m/s ² / $\sqrt{\text{Hz}}$]	[rad/s ² / $\sqrt{\text{Hz}}$]	[m/s ³ / $\sqrt{\text{Hz}}$]
EuRoC	ADIS16448	1.70×10^{-4}	2.00×10^{-3}	1.94×10^{-5}	3.00×10^{-3}
FIRoStereo	Epson G365	1.06×10^{-4}	8.56×10^{-4}	1.00×10^{-6}	1.00×10^{-4}
ROVTIO	VectorNav VN-100	2.80×10^{-4}	6.30×10^{-3}	6.20×10^{-5}	1.00×10^{-4}
STheReo	Xsens MTi-300	1.75×10^{-4}	5.89×10^{-4}	4.80×10^{-6}	1.47×10^{-4}

7 Results and Discussion

This chapter presents the per-dataset results and discusses them together, relating tracking performance to operating regime, preprocessing, and calibration, and answering the research questions.

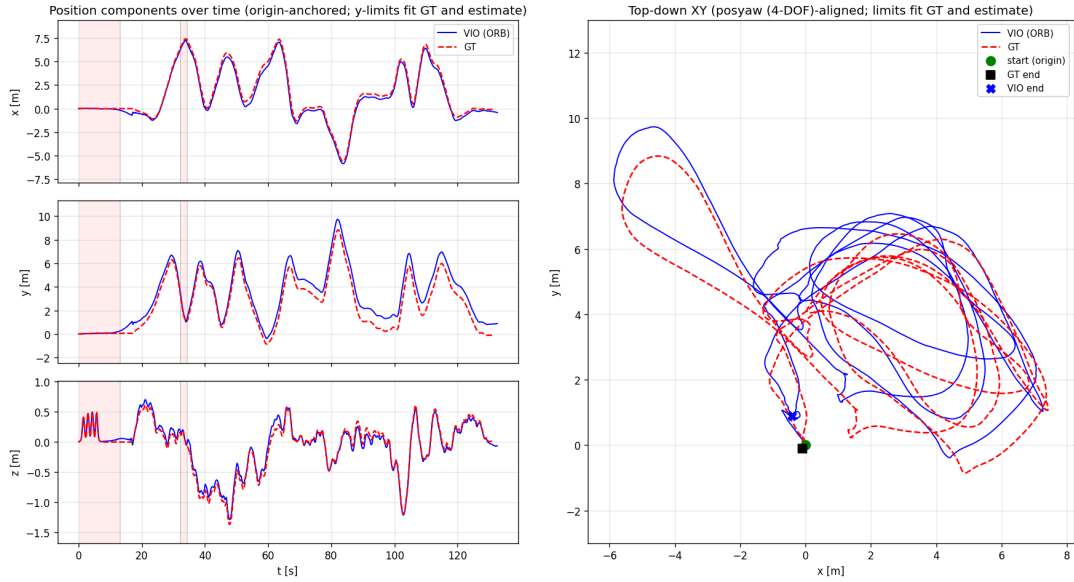
7.1 Evaluation Protocol

Each dataset is run in four configurations—the two front-ends (ORB, KLT) crossed with CLAHE off and on—under the shared parameter set of Section 5.7 and the adopted Norm→CLAHE preprocessing order (Section 7.3); the visible-light EuRoC sequence is included in two front-end configurations as a modality-independent baseline. Absolute Trajectory Error (ATE) is reported after 4-DOF (position and yaw) alignment, and Relative Pose Error (RPE) as the alignment-invariant per-metre translational error computed with `evo`. Because the pipeline exhibits a recurring transient–track–divergence pattern rather than a clean run, two ATE figures are distinguished: the full-trajectory ATE over the whole run, and the tracking-window ATE over the manually delimited interval in which the estimate follows the reference (the start-up transient and the post-divergence blow-up are excluded). The divergence and settling instants are read by hand off the position-versus-time plots, as the transient magnitudes on the thermal datasets (a 5–10 m start-up excursion before genuine tracking) make any fixed automatic threshold unreliable; each run additionally logs filter-consistency indicators (time-averaged normalised innovation NIS/df, and the used/in track ratio) and front-end track counts over time.

7.2 Baseline Validation on EuRoC

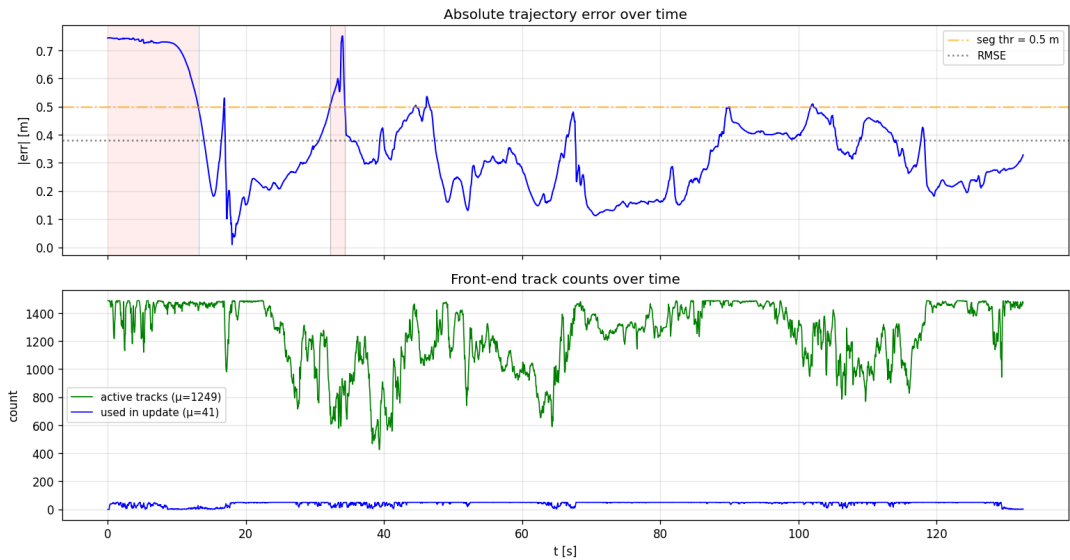
On the visible-light EuRoC MH_03 sequence the estimator recovers the metric trajectory across the whole run. With the ORB front-end it follows the reference from start to finish at a full-trajectory ATE of 0.38 m and a maximum instantaneous error of only 0.75 m (Figure 7.1), so scale, attitude and position are all recovered correctly—the strongest available evidence that the MSCKF back-end is sound. The KLT front-end behaves similarly but with a brief start-up transient: the drone is already manoeuvring at the start and the filter needs excitation before scale and velocity become observable, so it diverges momentarily and only locks on from roughly $t = 20$ s, which inflates its full-run ATE to 1.05 m; restricting the evaluation to the post-transient window brings this down to 0.46 m, though a residual excursion of up to 3 m remains on one axis. The apparent disagreement of the raw whole-run 4-DOF-aligned plot for KLT is largely this transient—aligning over the whole run lets the initial excursion rotate and translate the fit—and it shrinks once the alignment is restricted to the tracked interval. Either front-end confirms that the estimator can recover metric visual-inertial odometry when the modality is benign, so any degradation observed on the thermal datasets is attributable to the thermal modality (imagery and IMU parametrisation) rather than to an implementation fault, answering RQ1. The result does not by itself prove thermal viability, however, since EuRoC changes both the modality (visible vs. thermal) and the IMU quality simultaneously; it is a correctness check, not a like-for-like comparison.

euroc_mh03_ORB_CLAHEOFF [trajectory]



(a) Estimated trajectory against ground truth: position components over time and top-down XY.

euroc_mh03_ORB_CLAHEOFF [diagnostics]



(b) Diagnostics: absolute position error and front-end track counts over time.

Figure 7.1: EuRoC visible-light baseline (ORB, CLAHE off): the estimate follows the motion-capture reference for the whole run at a 0.38 m ATE, confirming back-end correctness (RQ1).

7.3 Preprocessing Order: CLAHE and Normalisation

The thermal front-end requires two operators that do not commute: contrast enhancement (CLAHE) and 8-bit normalisation. The order was selected empirically on the ROVTIO sequence, a well-conditioned indoor thermal case, before fixing it for the remaining datasets. Both orderings were run for each front-end (ORB, KLT) with CLAHE enabled; with CLAHE disabled the two orderings are identical, so a single CLAHE-off run per front-end is included as a reference. Because monocular scale is weakly observed on this indoor sequence, the absolute trajectory error (ATE) is dominated by scale error and is a misleading basis for ranking; the alignment-invariant relative pose error (RPE, per metre) is therefore taken as the primary criterion. Crucially, both metrics are computed over the same manually delimited tracking window (rather than the whole run), so that the ranking reflects tracking fidelity and is not distorted by the post-divergence blow-up.

Table 7.1: Preprocessing-order comparison on ROVTIO (a1t1), with *all* metrics computed over the tracking window. Tracking-window ATE and tracking-window RPE (evo, per metre); lower is better. RPE is the ranking criterion (alignment-invariant). CLAHE-off is shown as a reference (order not applicable). The lower on-run RPE per front-end is in bold.

Front-end	Preprocessing	track ATE [m]	RPE [m/m]
ORB	CLAHE-off (ref.)	1.18	0.45
ORB	CLAHE→Norm	1.98	0.61
ORB	Norm→CLAHE	4.20	0.64
KLT	CLAHE-off (ref.)	8.83	4.64
KLT	CLAHE→Norm	9.28	0.82
KLT	Norm→CLAHE	3.84	0.74

With every metric restricted to the tracking window, Table 7.1 tells a more nuanced story than a whole-run metric would. Within the interval in which the estimate actually tracks, the two orderings are nearly equivalent: for ORB the tracking-window RPE is 0.61 m/m for CLAHE→Norm against 0.64 m/m for Norm→CLAHE, a difference within noise, whereas for KLT the Norm→CLAHE ordering is the clearer winner (0.74 against 0.82 m/m). It is worth noting that the same comparison computed over the whole run would have shown a much larger—but misleading—gap for ORB: a full-run RPE of 1.26 m/m for Norm→CLAHE against 4.11 m/m for CLAHE→Norm. That apparent gap reflected the post-divergence blow-up rather than tracking quality and disappears once the metric is confined to the tracked segment, which is precisely why the tracking-window figures are used here. The ordering therefore has only a small, front-end-dependent effect on tracking fidelity. Because the two orderings perform so similarly within the tracked window, the choice is not expected to materially affect the results; Norm→CLAHE is nonetheless adopted for all subsequent experiments for three reasons: it is clearly the better option for KLT on both metrics (in-window RPE 0.74 against 0.82 m/m and tracking-window ATE 3.84 against 9.28 m) while remaining essentially neutral for ORB, it keeps the preprocessing consistent across datasets, and normalising the radiometric frame to 8-bit before enhancing its contrast is the conventional order for preparing thermal imagery. The effect is thus reported as marginal rather than decisive, and the discarded ordering is retained in the record for completeness.

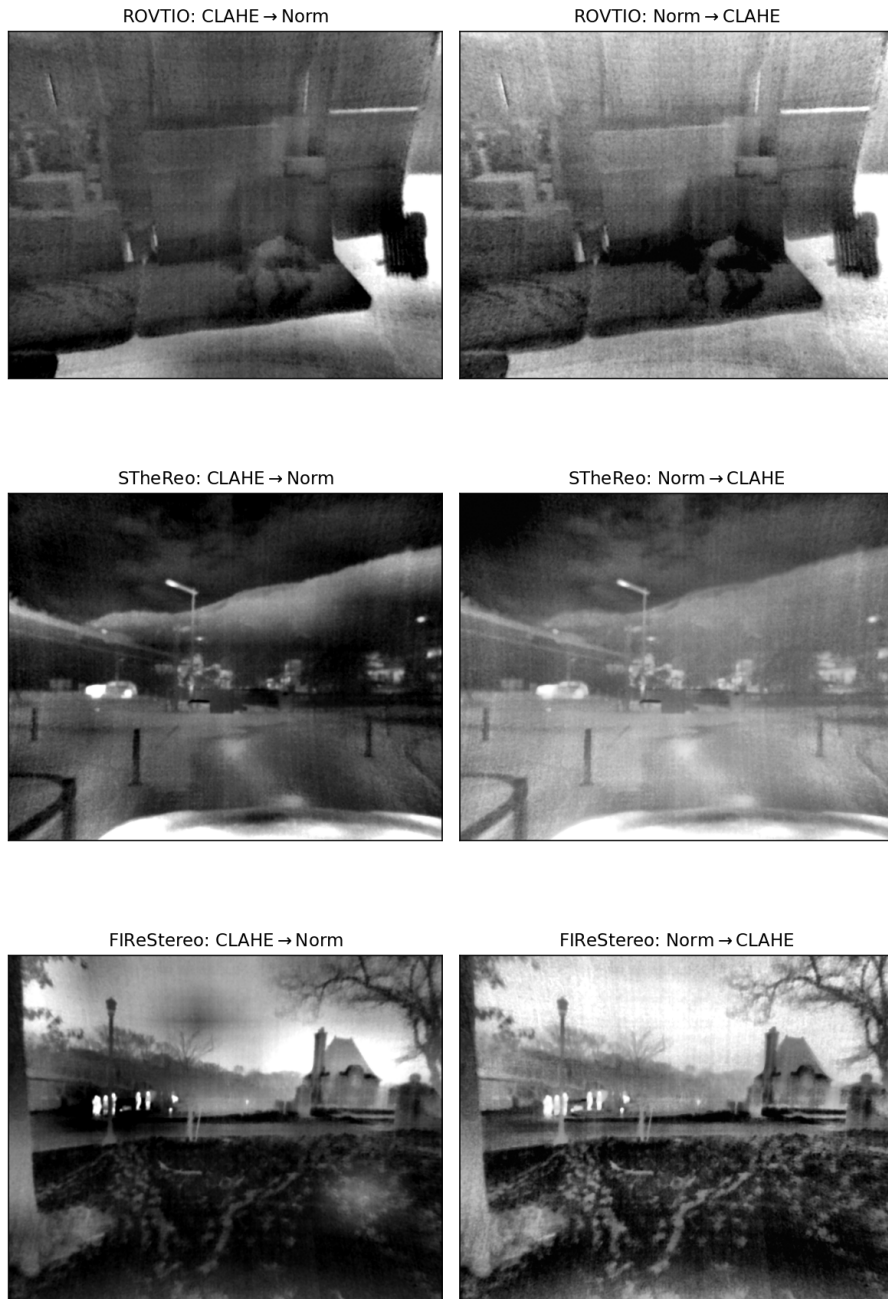
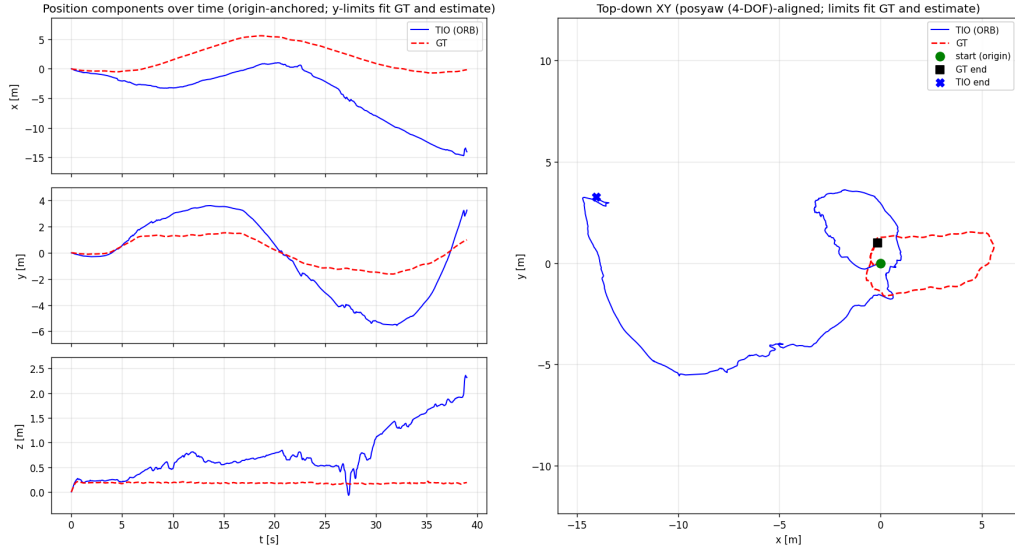


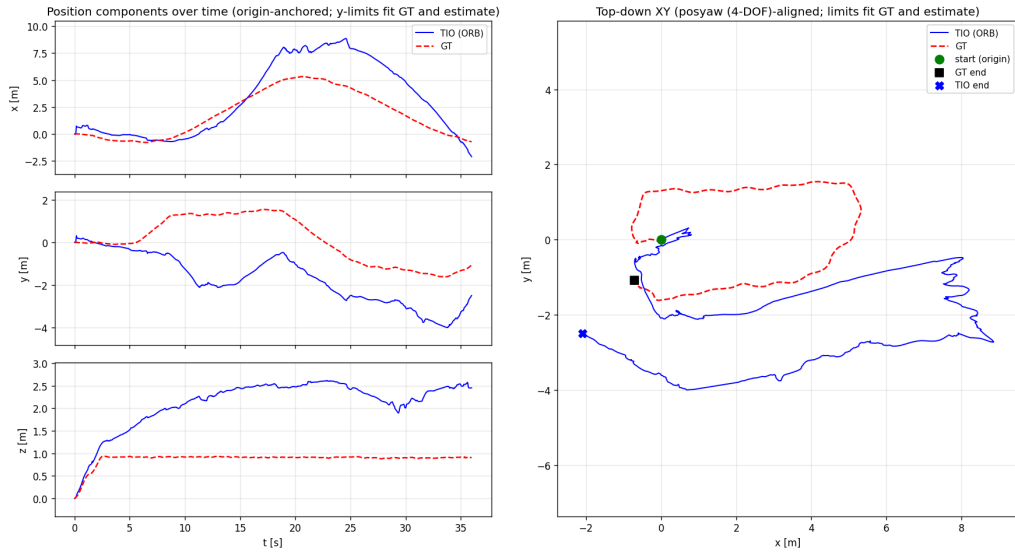
Figure 7.2: Preprocessing-order effect on the image (same middle frame from each thermal dataset, CLAHE enabled). Normalising to 8-bit *before* CLAHE (Norm $\rightarrow$ CLAHE, right column) preserves more local detail and avoids the highlight saturation that CLAHE $\rightarrow$ Norm (left column) produces in bright regions—most visible on FIReStereo, where the sky and foreground wash out under CLAHE $\rightarrow$ Norm. The visible difference is larger than the downstream effect on tracking, which is marginal (Table 7.1).

rovtio_alt1_ORB_CLAHEON_NORM-CLAHE [trajectory (up to divergence)]



(a) Norm→CLAHE (adopted): tracking-window trajectory, in-window RPE 0.64 m/m.

rovtio_alt1_ORB_CLAHEON_CLAHE-NORM [trajectory (up to divergence)]



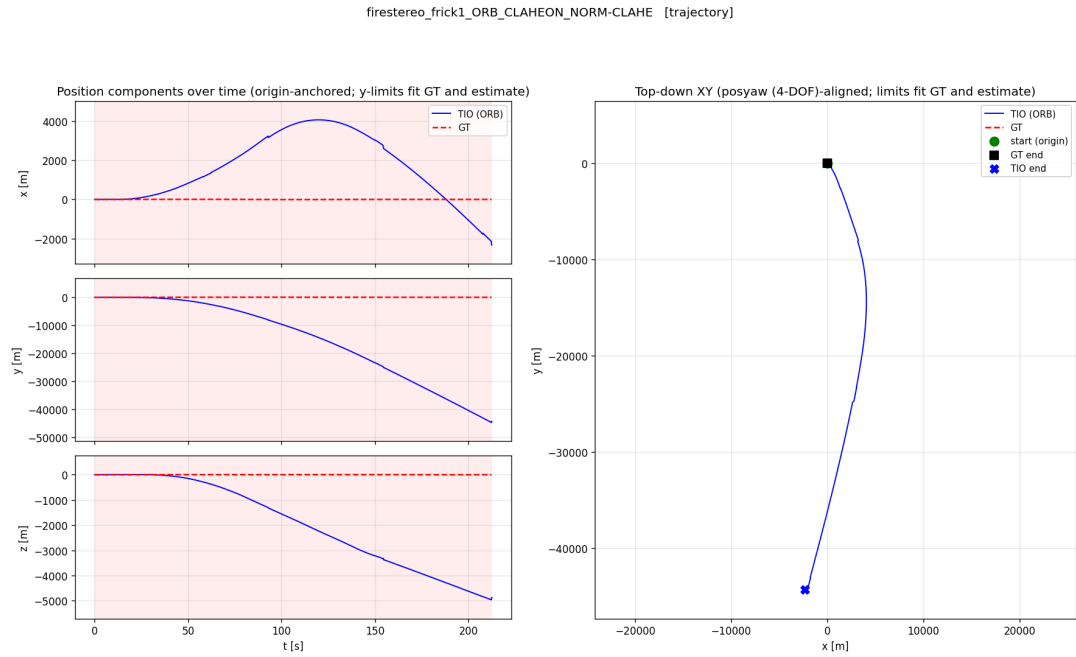
(b) CLAHE→Norm: tracking-window trajectory, in-window RPE 0.61 m/m.

Figure 7.3: Preprocessing-order effect on tracking (ROVTIO, ORB, CLAHE on): over the tracking window the two orderings follow the reference almost identically (in-window RPE 0.64 vs 0.61 m/m), consistent with the marginal ordering effect found in Table 7.1; the large whole-run RPE gap reported elsewhere reflects post-divergence behaviour rather than tracking quality.

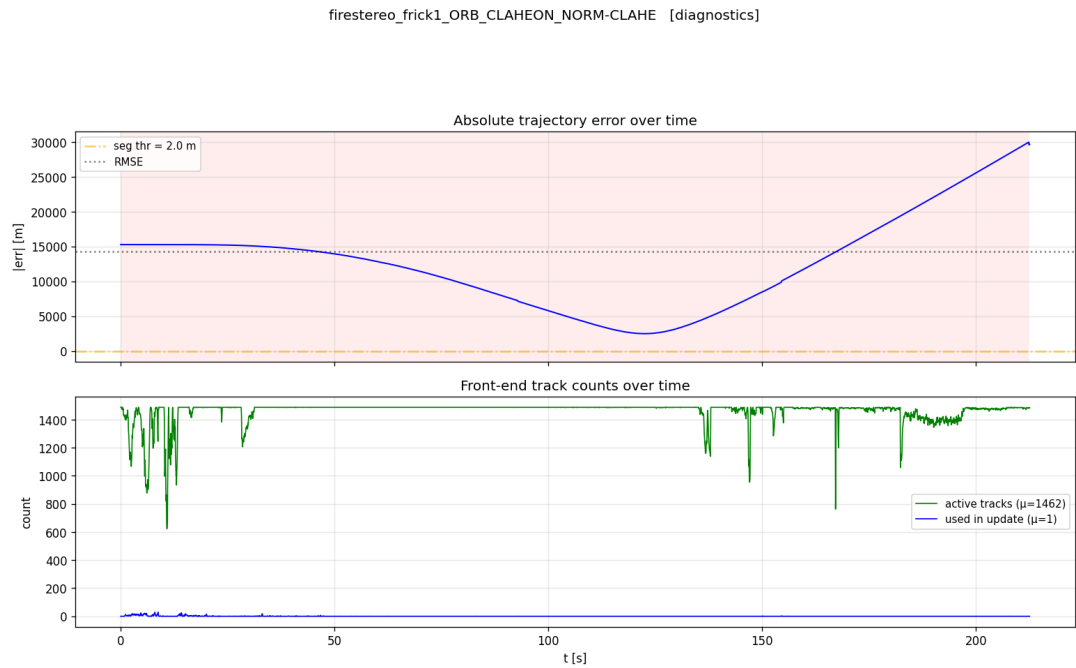
7.4 Per-Dataset Results

7.4.1 FIReStereo

FIReStereo does not track in any of the four configurations: across ORB/KLT and CLAHE off/on the mean number of tracks used per update is effectively zero (Table 7.2), at most one or two vertical-channel observations briefly agree with the reference at the very start, and the estimate is otherwise pure IMU dead-reckoning, giving trajectory errors of thousands of metres. The cause is geometric rather than photometric: the platform is a slow aerial system ($\lesssim 1$ m/s) observing a distant outdoor scene, so the inter-frame parallax rarely exceeds the minimum bearing-angle needed to triangulate a landmark, the triangulation and Gauss–Newton refinement reject nearly every track, and almost no measurement updates reach the filter. That CLAHE and the choice of front-end make no difference confirms the failure is not a preprocessing or matching issue—the front-end does produce matches—but a lack of usable parallax. FIReStereo is therefore reported as an operating-regime limit (Section 7.6): monocular thermal-inertial odometry is infeasible on slow, low-parallax platforms regardless of the image pipeline, and this run also exposes the cost of a single shared parameter set, since values that suit the faster platforms leave no headroom here.



(a) Estimated trajectory against reference: with almost no visual updates the estimate is effectively inertial dead-reckoning.



(b) Diagnostics: the used-track count stays near zero throughout, evidencing triangulation starvation.

Figure 7.4: FIREStereo (ORB, CLAHE on): too little parallax to triangulate, so almost no measurement updates occur and the estimate diverges as pure dead-reckoning; a regime limit rather than a preprocessing fault.

7.4.2 ROVTIO

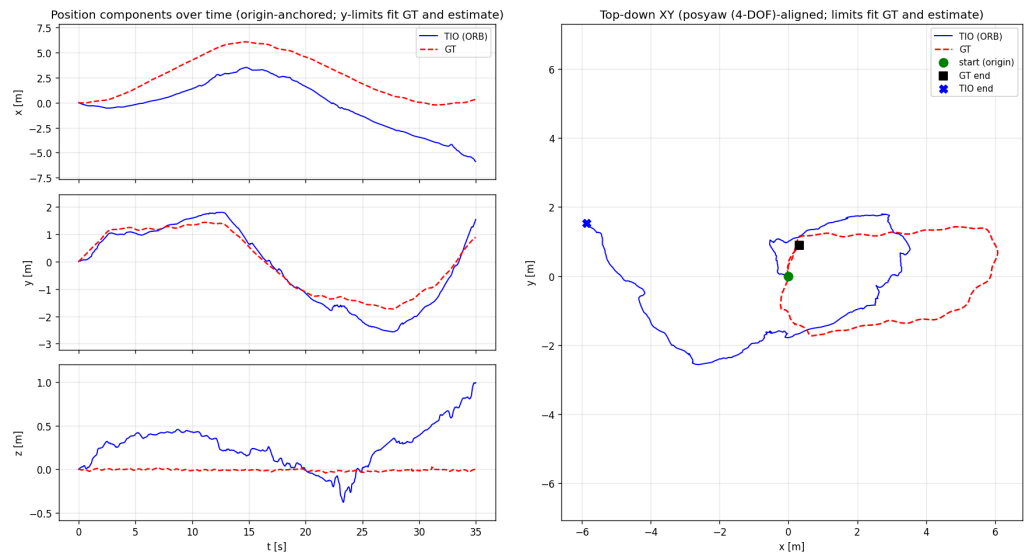
ROVTIO gives the most accurate absolute tracking of the thermal datasets—though not the longest-tracking one, since STheReo with KLT (Section 7.4.3) survives the whole run at a lower RPE. The indoor scene provides ample parallax and the platform starts from rest, so after a short start-up transient (roughly the first 14–20 s, during which the low-excitation start leaves scale and velocity weakly observable) the estimate begins to follow the reference and reproduces the same motion pattern for a further 30–40 s before diverging, typically around $t = 50\text{--}65$ s and usually just after a manoeuvre. The best configuration is ORB with CLAHE off, reaching a 1.18 m tracking-window ATE and a 0.45 m/m in-window RPE (the lowest in-window RPE among the thermal datasets); the adopted Norm→CLAHE runs are close behind (ORB 4.20 m / 0.64 m/m, KLT 3.84 m / 0.74 m/m), while KLT with CLAHE off nearly fails (8.83 m, in-window RPE 4.64, divergence by $t = 20$ s). This run never establishes a clean tracking phase, so its “tracking window” is really just the start-up transient and the approach to divergence; that is why, uniquely, its in-window RPE (4.64 m/m) slightly exceeds its full-run value (4.22 m/m), the windowed metric being meaningful only where genuine tracking occurs, as it does for every other run. A recurring feature is that the estimate keeps the correct *shape* of the trajectory while sitting at a large, roughly constant offset from the reference—the monocular scale is only weakly recovered on this indoor motion, so the absolute ATE is dominated by scale error even while the platform is genuinely being tracked. This is exactly the case that the absolute ATE penalises but the alignment-invariant RPE does not, which is why RPE (and the tracking-window ATE) are used to rank the runs; the full-trajectory ATE values of hundreds of metres reflect the post-divergence blow-up and the scale offset, not a total loss of tracking within the marked window.

rovtio_alt1_ORB_CLAHEOFF_CLAHE-NORM [trajectory]

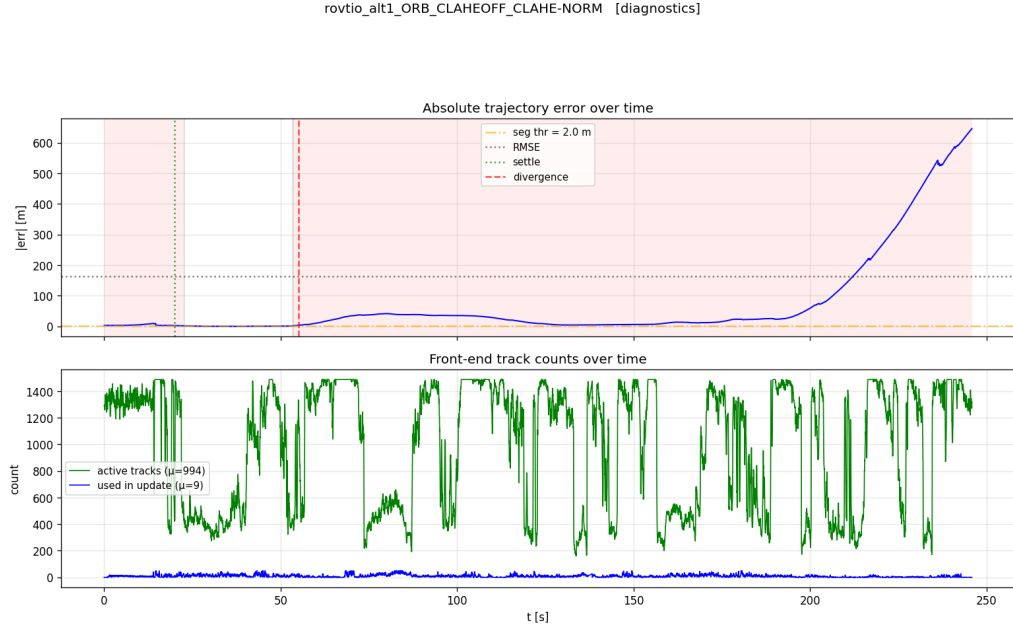


(a) Full run: tracks until $t \approx 55$ s then diverges, so the view is dominated by the blow-up.

rovtio_alt1_ORB_CLAHEOFF_CLAHE-NORM [trajectory (up to divergence)]



(b) Tracking window ([20, 55] s): follows the reference at a 1.18 m ATE, correct shape at a scale offset.



(c) Diagnostics: absolute position error and track counts over time.

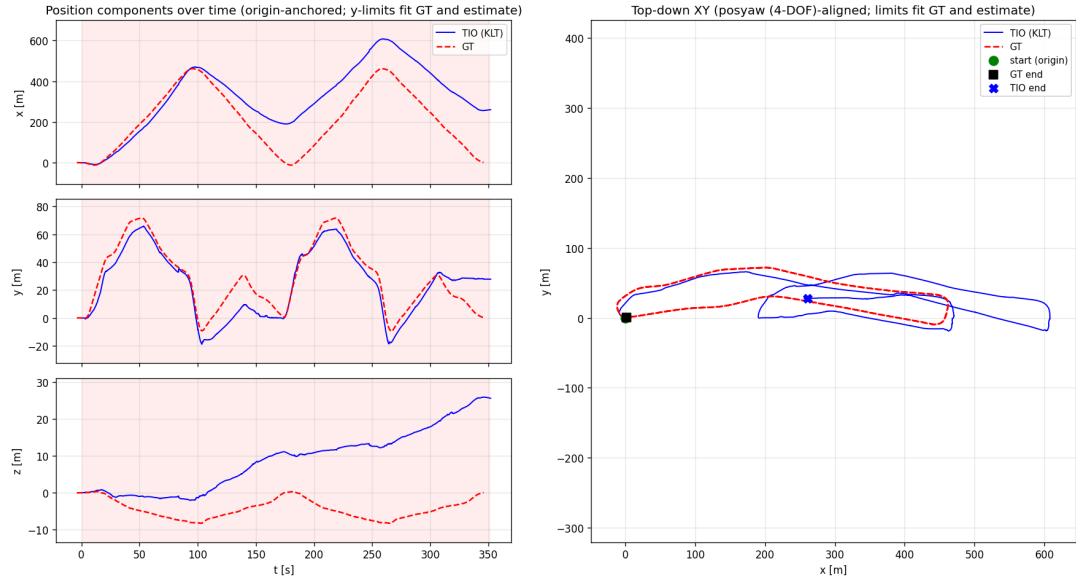
Figure 7.5: ROVTIO (ORB, CLAHE off): best in-window tracking of the thermal datasets; the accurate interval (b) is hidden in the full-run view (a) by the later divergence. Diagnostics in Figure 7.5c.

7.4.3 STheReo

STheReo is a car-mounted outdoor sequence whose difficulty is geometric rather than thermal: the forward-viewed scene is dominated by distant structure (a far tree-line and buildings) over a largely textureless near road, so usable parallax is scarce even though the vehicle moves quickly. The angular rates are in fact modest—peak ≈ 28 deg/s and 95th-percentile ≈ 17 deg/s—so the challenge is low parallax and the weak scale it produces, not fast rotation. Here the front-end choice is decisive: KLT with CLAHE on follows the trajectory for the entire run at the lowest RPE among the thermal datasets (0.77 m/m, tracking-window ATE effectively the full-run ATE of 96 m), keeping the motion shape from start to finish and never breaking on a turn; KLT with CLAHE off is similarly robust, surviving to $t = 270$ s at a somewhat higher whole-run RPE (1.11 m/m), consistent with CLAHE mitigating the brightness-constancy violation of Section 7.6 rather than being essential to survival. ORB, by contrast, is markedly more fragile—it loses tracking around the run’s sharpest turn and diverges by $t \approx 95$ s (38 m in-window ATE and an in-window RPE of 1.34 m/m, which its post-divergence tail inflates to 8.91 m/m over the whole run). The CLAHE-off ORB run shows the same post-turn fragility, though it survives somewhat longer (to $t \approx 120$ s) at a lower whole-run RPE (3.61 m/m); both ORB configurations remain far short of KLT, consistent with ORB being the weaker front-end on this low-parallax sequence. As with ROVTIO the error is scale-dominated (a persistent, growing offset from the reference while the direction of travel still agrees), and the vertical channel is briefly lost between $t \approx 50$ – 100 s before recovering. The failure mechanism on the ORB runs is instructive: the used-track count stays a small fraction of the active tracks throughout (used/in ≈ 0.20 against 0.58 for KLT) and almost every candidate is rejected by the Gauss–Newton triangulation stage. This is not an over-aggressive filter but the correct response to a low-parallax geometry—the distant, forward-viewed scene yields good two-dimensional matches yet too little parallax to triangulate depth, so the landmark solves to

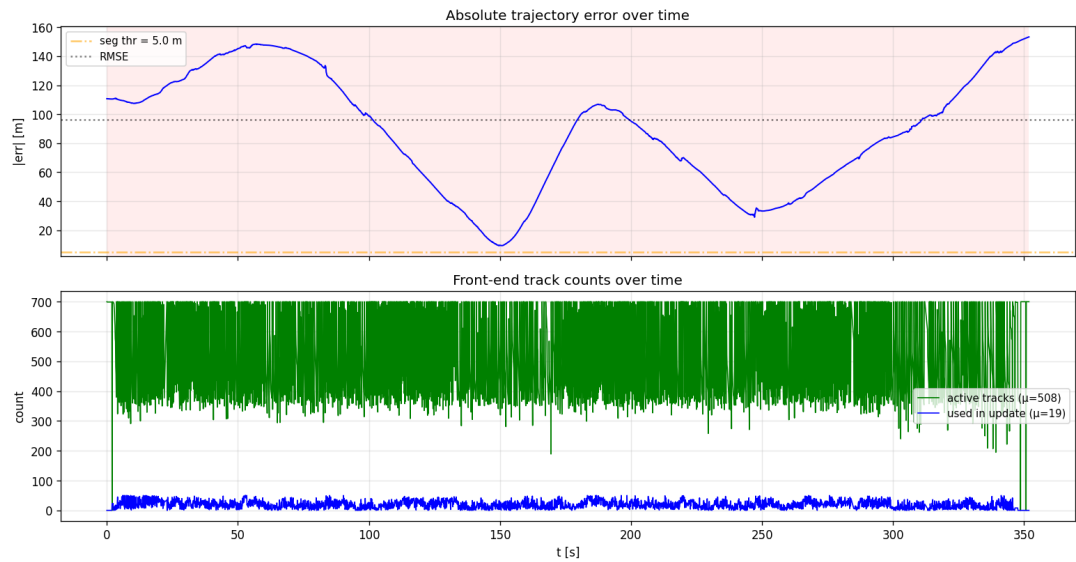
behind the camera or to a negative inverse depth and is dropped; admitting those depth-less points would inject garbage and diverge the filter faster. The scale-dominated error accumulates as a growing offset while the trajectory shape is still followed, and around the run's sharpest turn the combination tips the estimate into a divergence from which it does not recover (Figure 7.8). STheReo therefore marks the low-parallax, high-speed operating-regime limit for the fragile ORB front-end, survivable with the more robust KLT front-end (Section 7.6).

sthereo_valley_KLT_CLAHEON_NORM-CLAHE [trajectory]



(a) Estimated trajectory against ground truth: position components over time and top-down XY.

sthereo_valley_KLT_CLAHEON_NORM-CLAHE [diagnostics]



(b) Diagnostics: absolute position error and front-end track counts over time.

Figure 7.6: SThReo (KLT, CLAHE on): the estimate tracks the entire run at the lowest RPE among the thermal datasets (0.77 m/m), following the correct trajectory shape despite a large scale offset.

7.5 Cross-Dataset Analysis

Results are reported in two tables. Table 7.2 gives the full-run metrics over the entire recorded sequence, and Table 7.3 gives the metrics over only the manually delimited tracking window (from the settling instant, once the start-up transient has passed, to the divergence instant). Both use the adopted Norm→CLAHE preprocessing, with CLAHE off and on per front-end; for CLAHE off the ordering is inapplicable, so a single run is listed. ATE is the 4-DOF-aligned absolute error and RPE the alignment-invariant per-metre relative error (e_{vo}). Because monocular scale is only weakly observed, the absolute ATE is scale-dominated on the thermal sets—the estimate keeps the correct trajectory shape while sitting at a large offset—so the RPE, and especially the in-window RPE of Table 7.3, is the fairer indicator of tracking fidelity. Ground truth is external motion-capture for EuRoC and ROVTIO, and pseudo-ground-truth (LiDAR-inertial / local pose) for FIReStereo and STheReo.

Table 7.2: Full-run metrics (whole sequence, Norm→CLAHE). ATE (4-DOF) in metres; RPE in m/m; t_s = survival time to divergence in seconds (* = no divergence, tracked to the end); NIS/df is the time-averaged normalised innovation (≈ 1 ideal); used/in is the fraction of tracks passing the update gate. EuRoC is the visible-light baseline.

Dataset	FE	CLAHE	ATE	RPE	t_s [s]	NIS/df	used/in
ROVTIO	ORB	off	163	1.25	55	0.33	0.41
	ORB	on	256	1.26	55	0.34	0.45
	KLT	off	3169	4.22	20	0.72	0.15
	KLT	on	363	1.33	50	0.71	0.29
STheReo	ORB	off	889	3.61	120	0.31	0.46
	ORB	on	3500	8.91	95	0.26	0.20
	KLT	off	165	1.11	270	0.31	0.63
	KLT	on	96	0.77	352*	0.33	0.58
FIReStereo	ORB	on/off	<i>failed to track</i> (used/in ≈ 0 , no measurement updates)				
	KLT	on/off	<i>failed to track</i> (used/in ≈ 0 , ATE > 6000 m)				
EuRoC	ORB	off	0.4	0.12	133*	0.06	0.90
	KLT	off	1.1	0.18	133*	0.12	0.97

Table 7.3: Tracking-window metrics, over the manual interval [settle, divergence] only. ATE_{track} in metres; RPE_{track} in m/m (e_{vo} , computed on the window). FIReStereo has no tracking window (it never tracks).

Dataset	FE	CLAHE	window [s]	ATE_{track}	RPE_{track}
ROVTIO	ORB	off	[20, 55]	1.18	0.45
	ORB	on	[16, 55]	4.20	0.64
	KLT	off	[0, 20]	8.83	4.64
	KLT	on	[20, 50]	3.84	0.74
STheReo	ORB	off	[0, 120]	110.9	2.74
	ORB	on	[0, 95]	38.4	1.34
	KLT	off	[0, 270]	84.5	0.77
	KLT	on	[0, 352]	96.4	0.77
EuRoC	ORB	off	[0, 133]	0.38	0.12
	KLT	off	[20, 133]	0.46	0.09

The visible-light baseline confirms estimator correctness: once the start-up transient is excluded, EuRoC tracks the reference to a sub-metre ATE (0.38 m for ORB) and the lowest RPE of any run (0.12 m/m), and follows it for the whole run, so any thermal degra-

dation is attributable to the modality rather than an implementation fault (RQ1). That the visible-light baseline attains both the lowest ATE and the lowest RPE is a useful internal check: the RPE ranks the runs sensibly, which supports its use as the primary tracking-fidelity criterion. Two contrasting thermal successes emerge. STheReo with KLT is the most robust: it is the only thermal run that survives the entire sequence, at the lowest full-run RPE among the thermal datasets (0.77 m/m), despite a large scale offset that inflates its absolute ATE. ROVTIO with ORB is the most accurate *within its window*: while it tracks (a 35 s interval) it attains the lowest in-window RPE among the thermal datasets (0.45 m/m) and a 1.2 m ATE, but it then diverges near $t = 55$ s, which raises its full-run RPE to 1.25 m/m. Neither is unambiguously best—STheReo-KLT wins on robustness and full-run fidelity, ROVTIO-ORB on peak in-window accuracy—and the distinction is itself a finding: the absolute ATE is a poor cross-dataset ranking criterion here because it is dominated by scale, whereas the in-window RPE isolates true tracking quality. FIReStereo fails outright in every configuration: the slow aerial platform generates too little parallax to trigger triangulation, so almost no measurement updates occur—a regime limit, not a preprocessing fault (Section 7.6). Performance thus follows the operating regime: adequate parallax with moderate dynamics (ROVTIO) tracks accurately but briefly, marginal parallax from a distant forward-viewed scene at speed (STheReo) is survivable only with the more robust KLT front-end, and near-zero parallax from slow motion (FIReStereo) is infeasible monocularly (RQ2).

7.6 Failure-Mode Analysis

Three failure modes recur across the datasets, each supported by a specific piece of evidence rather than asserted. The first is *low-parallax triangulation starvation*, seen in its pure form on FIReStereo: the slow aerial platform never generates enough inter-frame bearing change to triangulate a landmark, so the parallax and Gauss–Newton stages reject almost every track and the used/in ratio stays near zero (Table 7.2); the front-end still produces matches, so this is a geometric limit of monocular observation, not a matching failure.

The second is *front-end-dependent triangulation starvation under low parallax*, seen on the STheReo ORB runs. The forward-viewed outdoor scene is dominated by distant structure over a largely textureless near road, so usable parallax is scarce; the more robust KLT front-end extracts enough to track the entire run, but ORB does not—its used-track count stays a small fraction of the active tracks throughout (Table 7.2, used/in 0.20 against 0.58 for KLT), because the depthless correspondences solve to a non-positive inverse depth or place the point behind the camera and the Gauss–Newton stage correctly discards them. The angular rates are modest (peak ≈ 28 deg/s), so this is a low-parallax rather than a fast-rotation degeneracy. The scale-dominated error therefore accumulates as a growing offset while the trajectory shape is still followed, until around the run’s sharpest turn (near $t = 95$ s) the combination tips the estimate into a divergence from which it does not recover (Figure 7.8); admitting the depthless tracks instead would inject inconsistent measurements and diverge the filter faster, so the rejection is protective, not the cause.

The third is *brightness-constancy violation from per-frame normalisation*, most visible on ROVTIO. The percentile normalisation is recomputed per frame, so abrupt scene-radiance changes (and the thermal core’s own automatic gain/uniformity correction) shift the mapped intensity of the same physical content substantially over a short time. Figure 7.7 quantifies this for a ROVTIO segment around $t = 67$ s, where the mapped intensity of the full frame drops by some 40 of 255 levels—and a fixed central patch swings by about 70—within roughly a second. This breaks the brightness-constancy assumption on which

direct KLT tracking depends, degrading the visual channel and contributing to the drift that precedes divergence; it is the reason the descriptor-based ORB front-end is competitive with, and on ROVTIO better than, KLT despite KLT's usual advantage on smooth motion.

Underlying all three is *weak scale observability*: metric scale is recovered only through acceleration excitation, so on the low-excitation starts and near-constant-velocity segments the scale is poorly constrained, which is why every thermal run keeps the correct trajectory shape while sitting at a large, growing offset from the reference (the scale-dominated ATE of Section 7.5). These mechanisms are not mutually exclusive: FReStereo is starved from the outset, STheReo survives until a manoeuvre collapses triangulation, and ROVTIO tracks briefly before the combination of brightness violation and weak scale drives it apart.

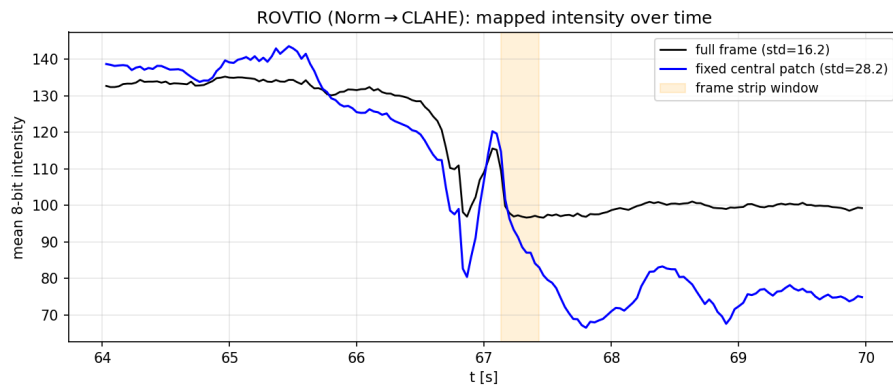
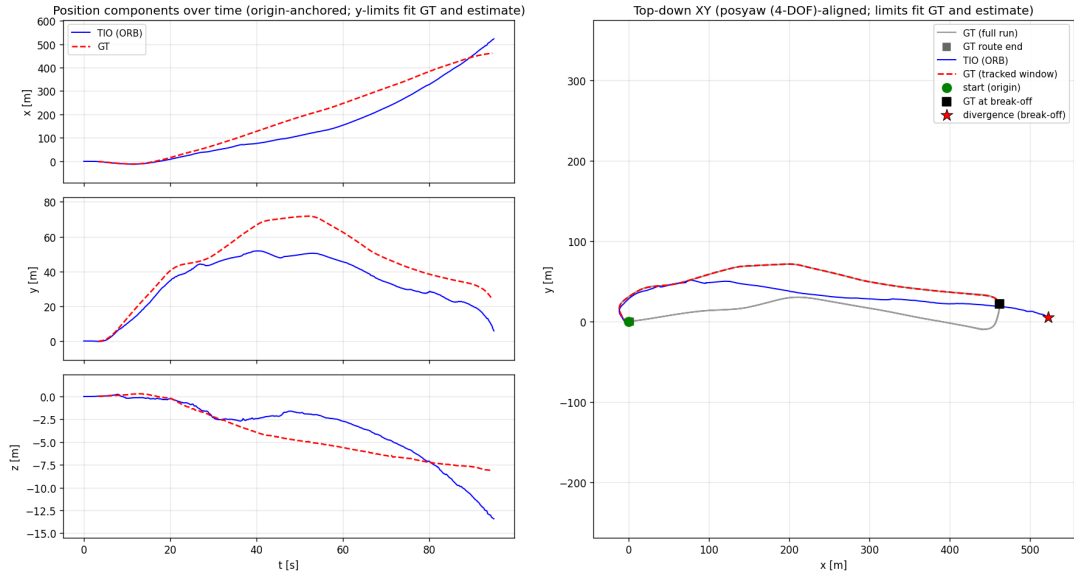
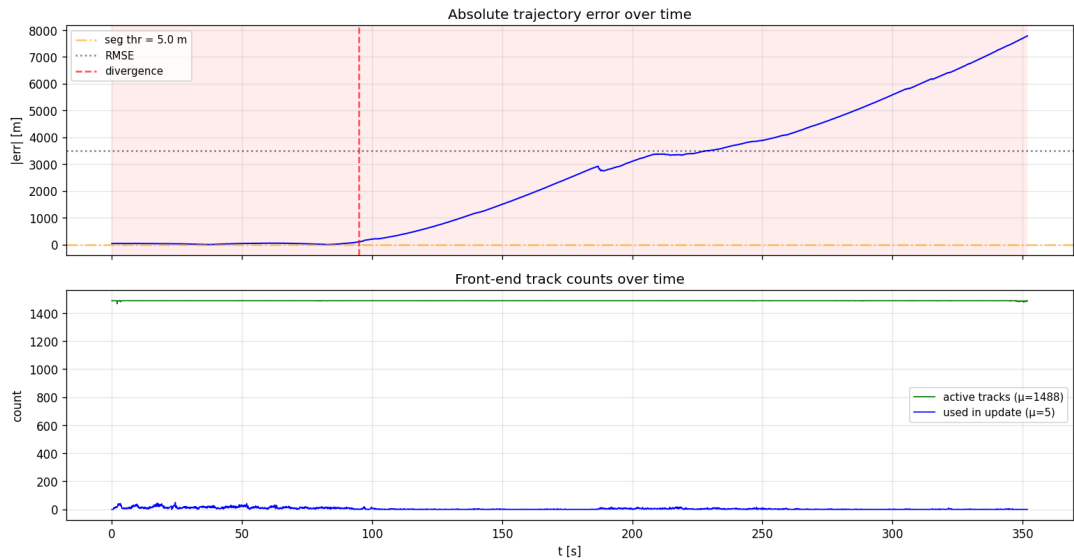


Figure 7.7: Brightness-constancy violation on ROVTIO (Norm→CLAHE). Mean mapped 8-bit intensity of the full frame (black) and of a fixed central image patch (blue) over time: around $t \approx 67$ s the full-frame intensity drops by ~ 40 levels and the central patch swings by ~ 70 (from 138 to 67) within about a second, so nominally static content is presented to the front-end at very different intensities. This violates the brightness-constancy assumption on which direct KLT tracking depends. The full-frame curve, a global statistic, is the cleaner indicator; the central-patch curve is more dramatic but partly confounded by camera motion over the window.



(a) Tracking window up to divergence, with the whole-run GT route (grey) for context: the estimate follows the true route at a growing scale offset, then breaks off (red star) around $t \approx 95$ s while GT continues (black square marks GT at that instant).



(b) Diagnostics: the absolute error is bounded while tracking, then grows without limit past $t \approx 95$ s; the used-track count stays near zero throughout (used/in ≈ 0.20), so the front-end is chronically starved.

Figure 7.8: STheReo ORB (CLAHE on) divergence: with scarce parallax the ORB front-end is chronically starved (b), so the scale-dominated error accumulates until, around the run's sharpest turn, the estimate breaks off the true route near $t \approx 95$ s and does not recover (a). The more robust KLT front-end survives the same sequence (Figure 7.6).

7.7 Sensitivity to Calibration

Intrinsics, camera–IMU extrinsics and the time offset are taken as provided by each dataset and held fixed (Section 3.4); the only sensor-model quantities varied are the IMU noise parameters, whose provenance differs by dataset (RQ3). EuRoC provides characterised values for its ADIS16448, whereas the thermal sets supply only datasheet figures for the Epson G365, VectorNav VN-100 and Xsens MTi-300; following the argument that the dataset-provided parametrisation is what should be evaluated, these are used directly, without the safety-margin inflation that a practitioner might otherwise apply, so that any inadequacy is exposed rather than masked. To separate the contribution of the inertial channel from that of the vision front-end, each platform is additionally run in a pure inertial dead-reckoning mode (no visual updates); Figure 7.9 overlays the resulting drift against time. This test characterises the *raw* inertial quality (bias and noise integrated open-loop) and is distinct from the noise-model adequacy probed by the consistency indicators of Section 7.8: the former asks how fast the IMU drifts on its own, the latter whether the covariance the filter assigns to that IMU is correct. The inertial-only drift separates the platforms into two groups (Figure 7.9). On ROVTIO and STheReo the open-loop drift is comparable to the characterised-IMU EuRoC baseline—under 25 m by $t = 30$ s—so for those sequences the divergences are not primarily inertial: the dominant error source is geometric (weak scale observability and low parallax), and the filter is not grossly over-confident (Section 7.8). FReStereo is the exception: its as-provided datasheet inertial model dead-reckons roughly an order of magnitude faster (about 200 m by $t = 30$ s against tens of metres for the others), so there a poor IMU parametrisation compounds the low-parallax vision starvation—exactly the datasheet inadequacy that the un-inflated, as-provided treatment was intended to expose. The overall picture is therefore that geometry dominates where the inertial channel is adequate (ROVTIO, STheReo), whereas on the slow aerial platform the vision and inertial channels fail together (RQ3).

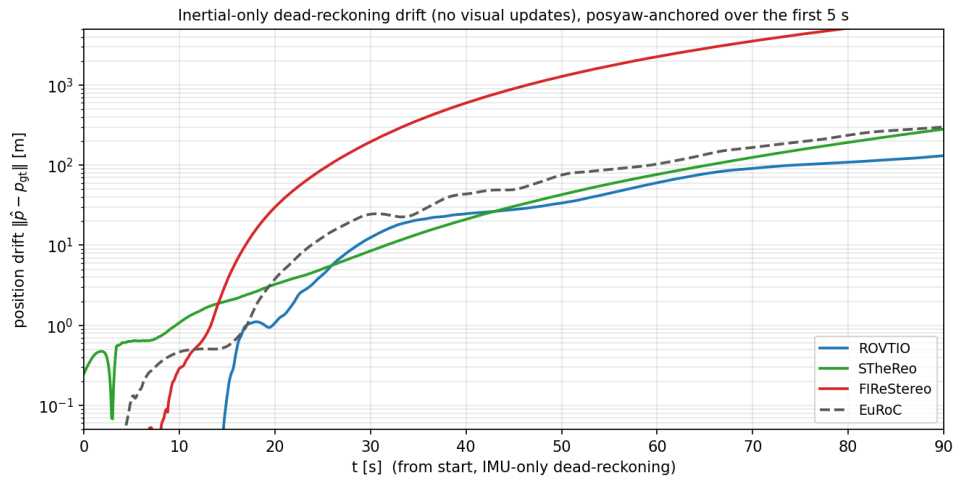


Figure 7.9: Inertial-only dead-reckoning drift for each platform (no visual updates), each aligned to GT with the 4-DOF (position and yaw) transform over the first 5 s and plotted on a logarithmic axis. ROVTIO, STheReo and the characterised-IMU EuRoC baseline drift comparably (tens of metres by $t = 30$ s), whereas FReStereo’s as-provided datasheet model drifts roughly an order of magnitude faster (≈ 200 m by $t = 30$ s); on that platform the inertial channel fails alongside the vision front-end, while on the others the geometry, not the IMU, dominates.

7.8 Filter Consistency

Filter consistency is monitored through the time-averaged normalised innovation squared per degree of freedom (NIS/df, ideally ≈ 1) and the used/in ratio of tracks passing the update gate (Table 7.2). Across all runs the reported NIS/df lies below one (0.06–0.72), which taken at face value indicates a *conservative* filter—innovations smaller than the assigned covariance predicts—rather than the over-confidence that a too-optimistic (un-inflated) IMU noise model might have been expected to produce. Two caveats temper this reading. First, the statistic is accumulated only over tracks that pass the chi-square gate, so it is truncated and biased downward; the true NIS/df is higher than reported and these values are best read as a lower bound, not as evidence of strong over-confidence. Second, the used/in ratio is low on the thermal sets (0.15–0.63 versus 0.90–0.97 on EuRoC), so the covariance is corrected by vision comparatively rarely and stays large, which is itself consistent with a below-one innovation ratio. The practical conclusion is that the divergences are not primarily a covariance mis-scaling problem: the filter is, if anything, under-confident, and the failures trace to the geometric mechanisms of Section 7.6 (starved or degenerate triangulation and weak scale) rather than to an over-trusted state. A rigorous consistency assessment (NIS against its chi-square bounds over Monte-Carlo repetitions, or NEES against ground truth) [17] is left as future work; only the aggregate indicator is reported here.

7.9 Limitations

Several limitations bound the strength of the conclusions and should be read alongside the results. The most fundamental is that the study is monocular by design: metric scale is observable only through inertial excitation, so the scale-dominated error that recurs across every thermal run is partly a consequence of the chosen configuration rather than of the thermal modality alone, and a stereo baseline would remove it. The evaluation also relies on a single shared parameter set across all datasets; this is deliberate, so that differences are attributable to the data rather than to per-dataset tuning, but it means the reported thermal accuracy is a lower bound—a regime-tuned configuration would almost certainly do better, most visibly on FIReStereo, where the shared settings leave no headroom.

The evaluation methodology carries its own caveats. Divergence and settling instants are marked by hand from the trajectory plots rather than detected automatically; this was a considered choice, because the transient-track-divergence pattern and the large start-up excursions defeat any fixed automatic threshold, but it introduces a degree of observer subjectivity and limits exact reproducibility. Each configuration is evaluated on a single sequence per dataset and from a single run, so run-to-run variability from the stochastic RANSAC stage is not quantified; the absolute figures should be read as representative rather than as statistically averaged. Ground truth is external motion-capture only for EuRoC and ROVTIO—for FIReStereo and STheReo it is a LiDAR-inertial or local-pose estimate with its own drift, so those errors are reported against a reference rather than against absolute truth. The filter-consistency indicator, likewise, is an aggregate NIS/df accumulated over gate-passing tracks and is therefore a truncated lower bound; a full consistency assessment (NIS against its chi-square bounds, or NEES against ground truth over repeated runs) is out of scope. Finally, the EuRoC baseline changes modality and IMU quality simultaneously and so validates estimator correctness without isolating the thermal contribution, the IMU noise parameters are taken from datasheets rather than Allan-variance characterisation, one candidate dataset (AerialTN) was excluded owing to an unresolved axis convention and a pre-compressed stream, and the pipeline is run offline with no compensation for motion blur or for the rolling-shutter readout of some

commercial thermal cores. None of these overturns the central finding, but each narrows the conditions under which it should be generalised.

8 Conclusion

This chapter answers the research questions, restates the contributions in light of the evidence, and outlines directions for future work.

8.1 Summary of Findings

On RQ1 (feasibility and baseline), the same MSCKF estimator that recovers the EuRoC visible-light trajectory to a sub-metre error over the whole run does produce metric odometry from raw thermal imagery and a low-cost IMU—but only briefly and only within favourable regimes. The accuracy gap to the visible-light baseline is large: where EuRoC tracks the entire sequence, the thermal runs keep the correct trajectory shape at a poorly-recovered scale and, in most configurations, diverge partway through. The baseline is best read as a correctness check on the estimator rather than as proof of thermal viability, since it changes both the modality and the IMU quality at once.

On RQ2 (operating regime), performance follows the motion and scene geometry far more strongly than the thermal contrast. Adequate parallax with moderate dynamics (ROV-TIO, indoor) yields the most accurate in-window tracking; marginal parallax from a distant forward-viewed scene at speed (STheReo, vehicle) is survivable only with the more robust KLT front-end, since ORB is starved of usable parallax and the run’s sharpest turn tips its accumulated error into an unrecoverable divergence; and a slow, low-parallax platform (FIReStereo) is infeasible monocularly, generating too few triangulable landmarks for any visual update. The recurring failure signature is triangulation starvation under low parallax—chronic for the slow platform, front-end-dependent for the vehicle—together with weak scale observability under low acceleration excitation.

On RQ3 (preprocessing and IMU-noise parametrisation), the ordering of the thermal preprocessing operators has a small but consistent effect—Norm→CLAHE gives the lower RPE and was adopted—while per-frame percentile normalisation introduces a genuine brightness-constancy violation that degrades the direct KLT front-end and helps explain why the descriptor-based ORB is competitive on thermal data. The IMU-noise parametrisation, by contrast, is *not* the dominant error source where the inertial channel is adequate: on ROVTIO and STheReo the open-loop inertial drift matches the characterised-IMU baseline and the filter is if anything under-confident (NIS/df below one, albeit a truncated lower bound), so the divergences there trace to the geometric mechanisms above rather than to a mis-scaled inertial covariance. The clear exception is FIReStereo, whose as-provided datasheet model dead-reckons roughly an order of magnitude faster, so its failure is doubly determined—negligible parallax for the vision channel and a poor inertial model—which is precisely the datasheet inadequacy the un-inflated treatment set out to expose. The overall conclusion is that classical monocular thermal-inertial odometry is viable only within a narrow band of favourable operating conditions and is not, on its own, a reliable metric odometry source under unconstrained motion.

8.2 Contributions Revisited

The four stated contributions hold up against the evidence, with the emphasis shifted by what the experiments revealed. The modular pipeline (decoupled preprocessing, KLT/ORB front-end, MSCKF back-end) made it possible to attribute each failure to a specific layer—front-end starvation, triangulation geometry, or brightness violation—rather than to the

system as a whole. The consistent multi-dataset protocol, running an identical configuration across four datasets, is what exposed the operating-regime dependence so clearly; it also carried a cost, since the single shared parameter set left no headroom on the slow FIReStereo platform, which is itself an informative negative result. The justified 4-DOF (position and yaw) evaluation methodology proved essential rather than cosmetic: the absolute ATE turned out to be dominated by unrecovered scale, so an alignment-invariant relative metric was needed to rank tracking fidelity fairly, and the tracking-window RPE became the study’s decisive indicator. Finally, the sensitivity and failure-mode analysis delivered concrete, evidence-linked mechanisms—low-parallax triangulation starvation, post-manoeuvre triangulation collapse, brightness-constancy violation, and weak scale observability—each tied to a measurable signal (used/in ratio, track-count collapse, per-frame intensity shift) rather than asserted.

8.3 Future Work

Several extensions follow directly from the limitations identified above. The most immediate is to replace the single shared configuration with *regime-aware parameter selection*: optimising the filter and preprocessing parameters and tuning them per dataset or per motion characteristic (platform speed, expected parallax, contrast) rather than fixing one compromise, which the FIReStereo failure shows is necessary for slow, low-parallax platforms. A second direction is a richer front-end. Thermal scenes are often low-textured and corner-poor, so a point-and-line formulation in the spirit of PL-VIO [9]—tracking edges in addition to corners—should provide constraints where sparse corner tracking starves, and is a natural fit for the man-made and structured environments where thermal odometry is most useful. Third, the confounding between modality and IMU quality that limits the EuRoC comparison motivates *a purpose-built dataset* for this problem: thermal imagery delivered as raw high-bit-depth frames together with an Allan-variance-characterised IMU, which would let the thermal modality and the inertial parametrisation be varied independently. Fourth, and most decisive for the core weakness, a *stereo thermal* configuration would make metric scale directly observable from the fixed baseline rather than relying on inertial excitation, addressing the scale-dominated error that recurs throughout this study; sensor fusion with an additional modality (for example a range or altitude sensor, or radar as demonstrated in [15]) is an alternative route to the same end. Beyond these, a learning-based or hybrid front-end robust to thermal artefacts, rolling-shutter and motion-blur compensation for commercial thermal cores, and real-time on-board deployment remain worthwhile but secondary to resolving scale and front-end starvation first.

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A Appendix

This appendix collects two kinds of supplementary material: Section A.1 gives the explicit noise-input matrix $\mathbf{G}$ and discretised process-noise covariance $\mathbf{Q}$ referenced from Section 4.2.3, and Sections A.2–A.4 supply trajectory and diagnostics figures for run configurations that are reported only as numbers in Chapter 7.

A.1 Error-State Propagation Noise: Construction of $\mathbf{G}$ and $\mathbf{Q}$

Section 4.2.3 gives the continuous error-state dynamics and states that the white-noise terms $(\mathbf{n}_g, \mathbf{n}_a, \mathbf{n}_{bg}, \mathbf{n}_{ba})$ enter through a noise-input matrix $\mathbf{G}$, which together with their spectral densities defines the discrete process-noise covariance $\mathbf{Q}$. This section makes that construction explicit, using the error-state ordering $\tilde{\mathbf{x}}_I = [\delta\boldsymbol{\theta}^\top, \delta\mathbf{b}_g^\top, \delta\mathbf{v}^\top, \delta\mathbf{b}_a^\top, \delta\mathbf{p}^\top]^\top$ already declared in Eq. 4.6, which pairs each error component with the bias driving it and matches the implementation directly.

Collecting the noise terms of Eqs. 4.6–4.5 into $\mathbf{n} = [\mathbf{n}_g^\top, \mathbf{n}_a^\top, \mathbf{n}_{bg}^\top, \mathbf{n}_{ba}^\top]^\top \in \mathbb{R}^{12}$, the continuous error dynamics can be written $\dot{\tilde{\mathbf{x}}}_I = \mathbf{F}\tilde{\mathbf{x}}_I + \mathbf{G}\mathbf{n}$, with the 15×12 noise-input matrix

$$\mathbf{G} = \begin{bmatrix} -\mathbf{I}_3 & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{I}_3 & \mathbf{0} \\ \mathbf{0} & -\mathbf{R}_I^W & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{I}_3 \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \end{bmatrix}, \quad (\text{A.1})$$

whose row blocks follow the state order above and whose column blocks follow $\mathbf{n}$: the $-\mathbf{I}_3$ and $\mathbf{I}_3$ blocks read off the noise terms directly attached to $\delta\boldsymbol{\theta}$, $\delta\mathbf{b}_g$, $\delta\mathbf{b}_a$ in Section 4.2.3; the $-\mathbf{R}_I^W$ block does the same for $\delta\mathbf{v}$; and the position row is zero because $\delta\mathbf{p} = \delta\mathbf{v}$ has no direct noise term, only the noise it inherits from $\delta\mathbf{v}$ through propagation.

The four noise processes are modelled as mutually independent white noise with constant continuous-time spectral densities $\sigma_g^2, \sigma_a^2, \sigma_{bg}^2, \sigma_{ba}^2$ (the gyroscope and accelerometer noise densities and bias random-walk rates of Section 5.7, whose dataset-specific values are given in Chapter 6), so $\mathbf{n}$ has spectral density $\mathbf{Q}_c^{\text{raw}} = \text{blockdiag}(\sigma_g^2\mathbf{I}_3, \sigma_a^2\mathbf{I}_3, \sigma_{bg}^2\mathbf{I}_3, \sigma_{ba}^2\mathbf{I}_3)$. The continuous-time process-noise covariance of the error state is then $\mathbf{Q}_c = \mathbf{G}\mathbf{Q}_c^{\text{raw}}\mathbf{G}^\top$, which block-diagonalises because each row block of $\mathbf{G}$ has non-zero entries in only one noise column:

$$\mathbf{Q}_c = \text{blockdiag}(\sigma_g^2\mathbf{I}_3, \sigma_{bg}^2\mathbf{I}_3, \sigma_a^2(\mathbf{R}_I^W)(\mathbf{R}_I^W)^\top, \sigma_{ba}^2\mathbf{I}_3, \mathbf{0}). \quad (\text{A.2})$$

Because $\mathbf{R}_I^W$ is orthogonal, $\mathbf{R}_I^W(\mathbf{R}_I^W)^\top = \mathbf{I}_3$, so the velocity block reduces to $\sigma_a^2\mathbf{I}_3$ regardless of the current attitude: an isotropic (rotationally invariant) noise spectral density in the body frame remains isotropic in the world frame under an orthogonal rotation. Equation A.2 therefore simplifies to the attitude-independent diagonal

$$\mathbf{Q}_c = \text{diag}(\sigma_g^2, \sigma_g^2, \sigma_g^2, \sigma_{bg}^2, \sigma_{bg}^2, \sigma_{bg}^2, \sigma_a^2, \sigma_a^2, \sigma_a^2, \sigma_{ba}^2, \sigma_{ba}^2, \sigma_{ba}^2, 0, 0, 0), \quad (\text{A.3})$$

which is exactly the fixed matrix built once from the dataset’s noise parameters and held constant across the run, since it depends on the noise densities alone and not on the current state estimate.

Given the first-order discrete transition $\Phi = \mathbf{I} + \mathbf{F}\Delta t$ of Section 4.2.3, the exact discrete process-noise covariance over one propagation interval is the Van Loan integral $\mathbf{Q}_d = \int_0^{\Delta t} \Phi(\tau) \mathbf{Q}_c \Phi(\tau)^\top d\tau$. At the IMU rate used here ($\Delta t \approx 0.01$ s), the zeroth-order term dominates and the implementation retains only that term,

$$\mathbf{Q}_d \approx \mathbf{Q}_c \Delta t, \quad (\text{A.4})$$

consistent with, and at the same order as, the first-order truncation already used for Φ . The terms this omits are $O(\Delta t^2)$ corrections that couple the noise entering through $\delta \mathbf{v}$ into the $\delta \mathbf{p}$ – $\delta \mathbf{v}$ covariance block (from $\mathbf{F}\mathbf{Q}_c + \mathbf{Q}_c\mathbf{F}^\top$) and an $O(\Delta t^3)$ term for the $\delta \mathbf{p}$ – $\delta \mathbf{p}$ block; at 100 Hz these are negligible next to the zeroth-order term and are not implemented.

A.2 EuRoC

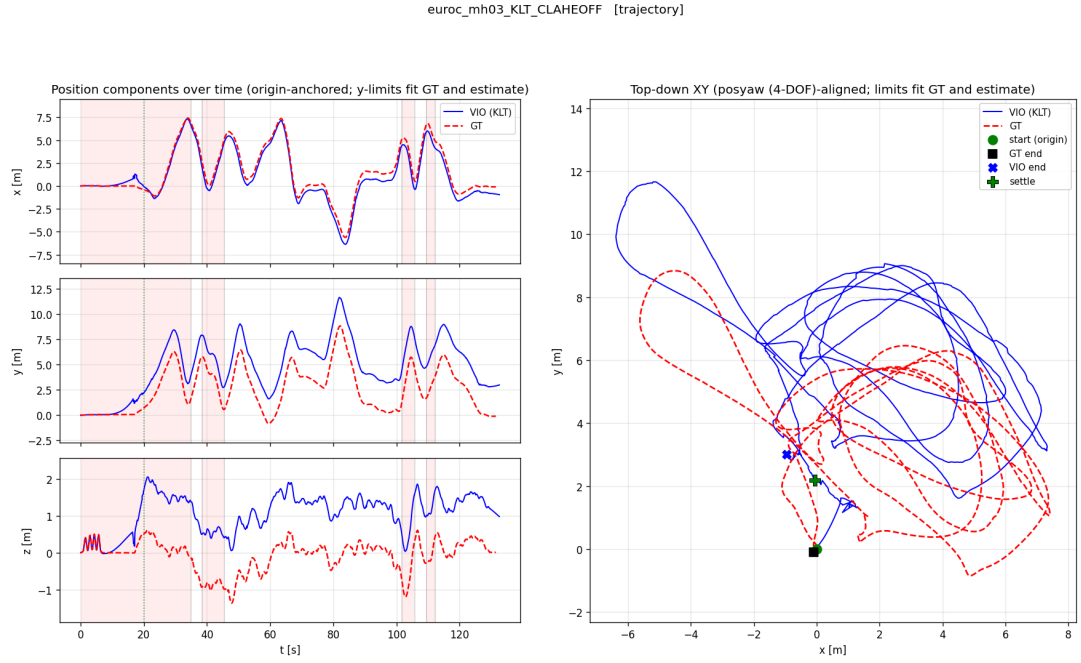
The KLT configuration (Figure A.1) complements the ORB baseline of Section 7.2: after the start-up transient described there, it tracks the reference for the remainder of the run.

A.3 ROVTIO

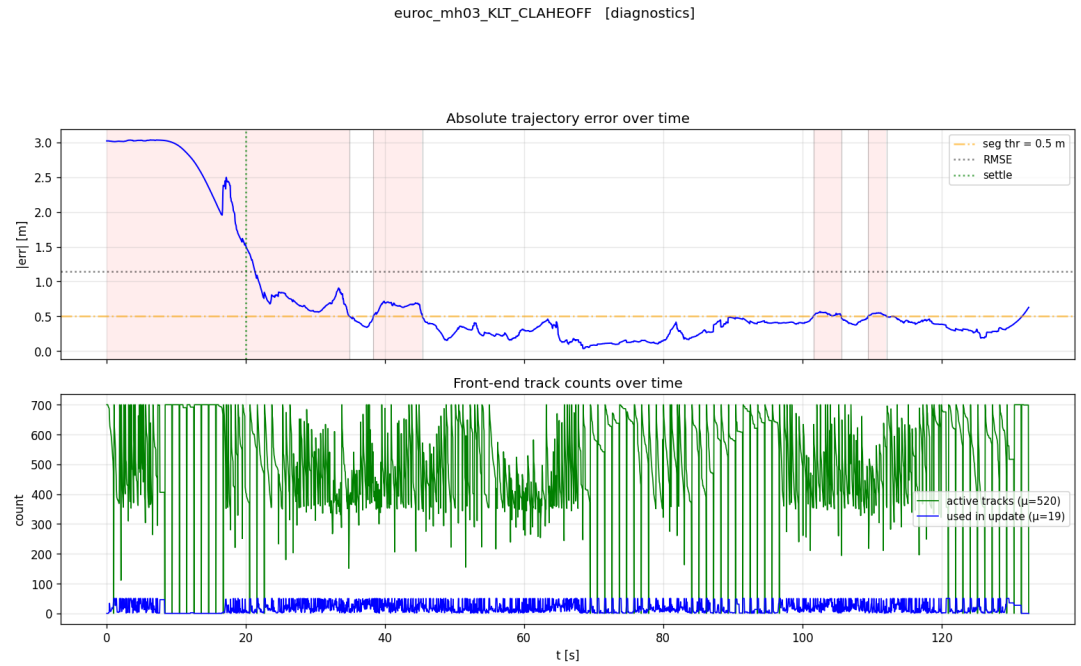
Figure A.2 shows the KLT, CLAHE-off configuration referred to in Section 7.4.2 as “nearly failing”: tracking window $[0, 20]$ s, $\text{ATE}_{\text{track}} = 8.83$ m, $\text{RPE}_{\text{track}} = 4.64$ m/m. Figure A.3 shows the KLT counterpart of the adopted Norm→CLAHE ordering (CLAHE on), which tracks somewhat longer and considerably more accurately (window $[20, 50]$ s, $\text{ATE}_{\text{track}} = 3.84$ m, $\text{RPE}_{\text{track}} = 0.74$ m/m).

A.4 STheReo

Figures A.4 and A.5 show the CLAHE-off counterparts of the ORB and KLT runs discussed in Section 7.4.3. Both survive substantially longer than the CLAHE-on ORB run (to $t = 120$ s and $t = 270$ s respectively), consistent with CLAHE mitigating rather than causing the fragility discussed there.



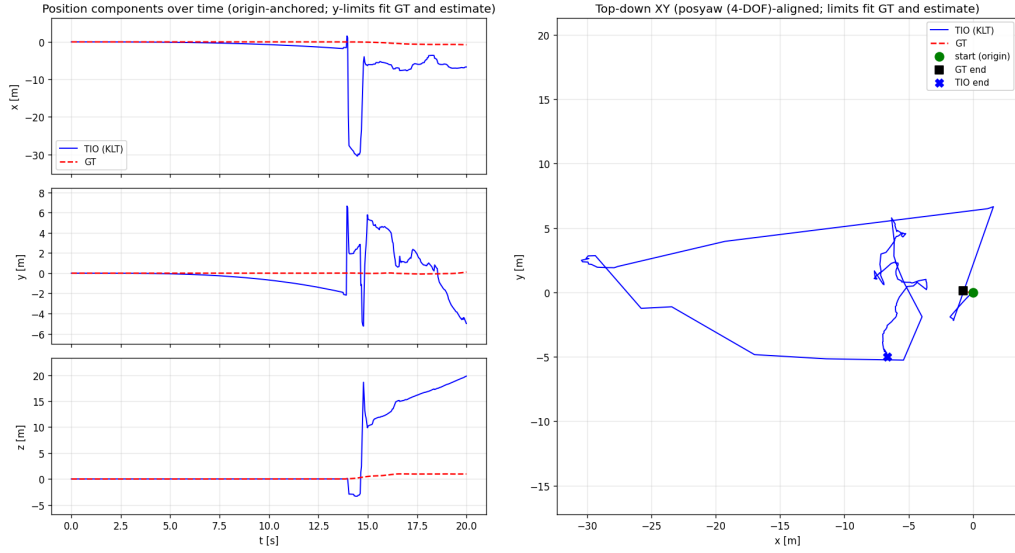
(a) Estimated trajectory against ground truth: position components over time and top-down XY.



(b) Diagnostics: absolute position error and front-end track counts over time.

Figure A.1: EuRoC (KLT, CLAHE off): after the start-up transient (Section 7.2) the estimate locks onto the reference and tracks it for the remainder of the run.

rovtio_alt1_KLT_CLAHEOFF_CLAHE-NORM [trajectory (up to divergence)]



(a) Tracking window $[0, 20]$ s.

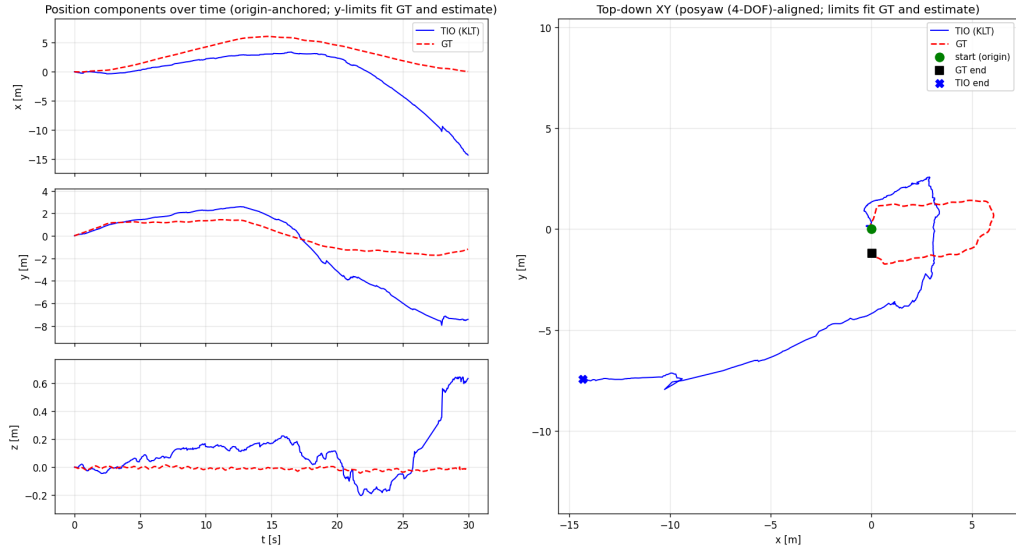
rovtio_alt1_KLT_CLAHEOFF_CLAHE-NORM [diagnostics]



(b) Diagnostics: absolute position error and track counts over time.

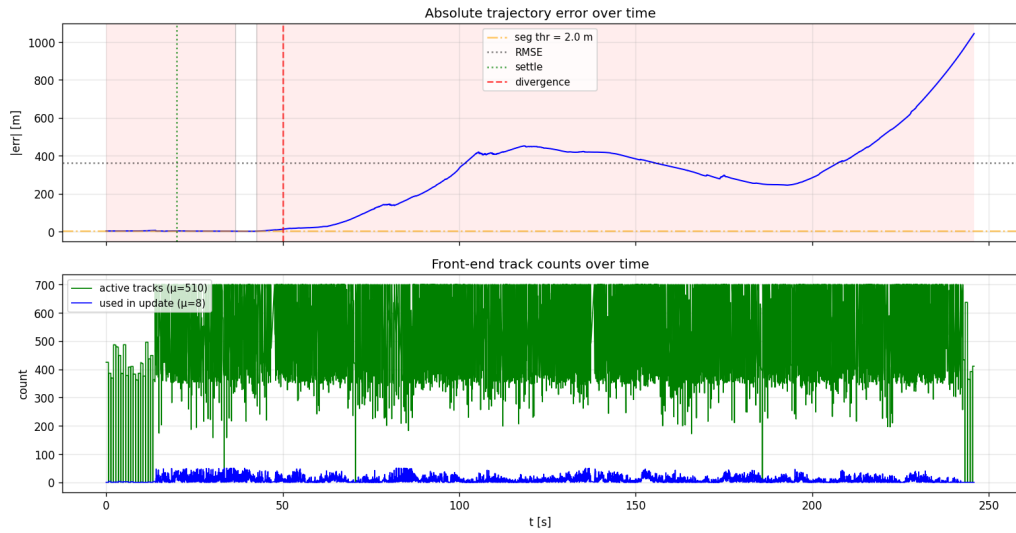
Figure A.2: ROVTIO (KLT, CLAHE off): the weakest of the four ROVTIO configurations, diverging by $t = 20$ s.

rovtio_alt1_KLT_CLAHEON_NORM-CLAHE [trajectory (up to divergence)]



(a) Tracking window [20, 50] s.

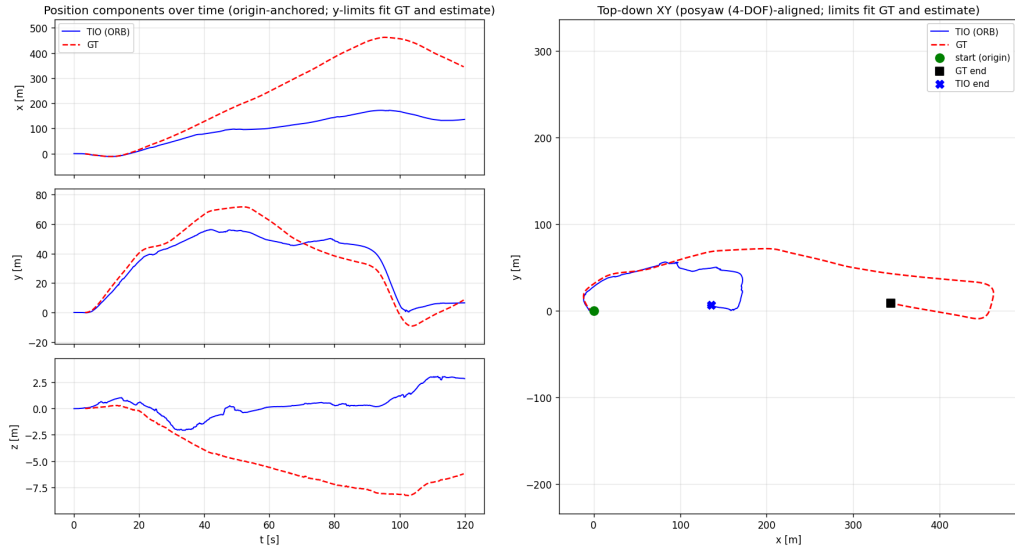
rovtio_alt1_KLT_CLAHEON_NORM-CLAHE [diagnostics]



(b) Diagnostics: absolute position error and track counts over time.

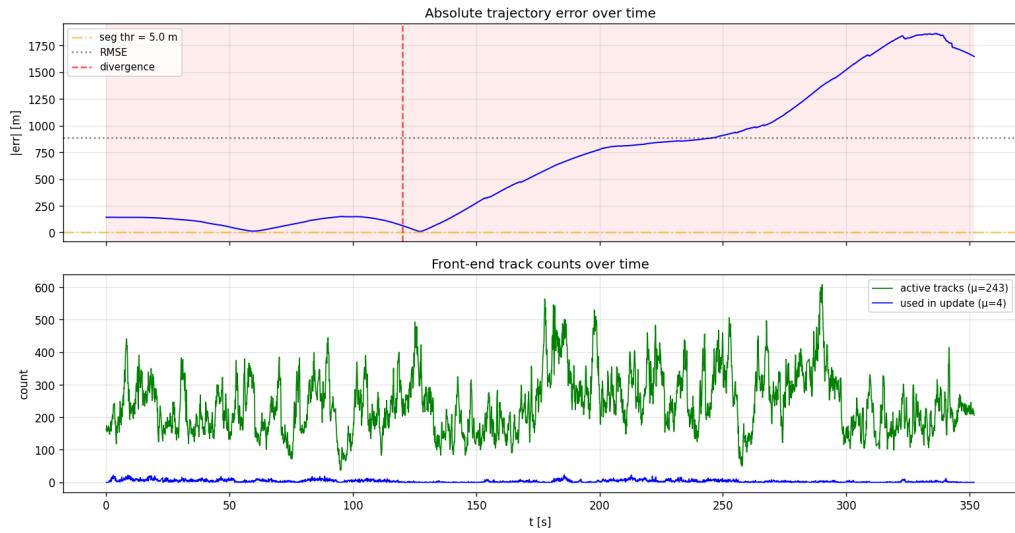
Figure A.3: ROVTIO (KLT, CLAHE on, Norm→CLAHE): the adopted-ordering KLT run, close behind the best ORB configuration of Section 7.4.2.

stereo_valley_ORB_CLAHEOFF_NORM-CLAHE [trajectory (up to divergence)]



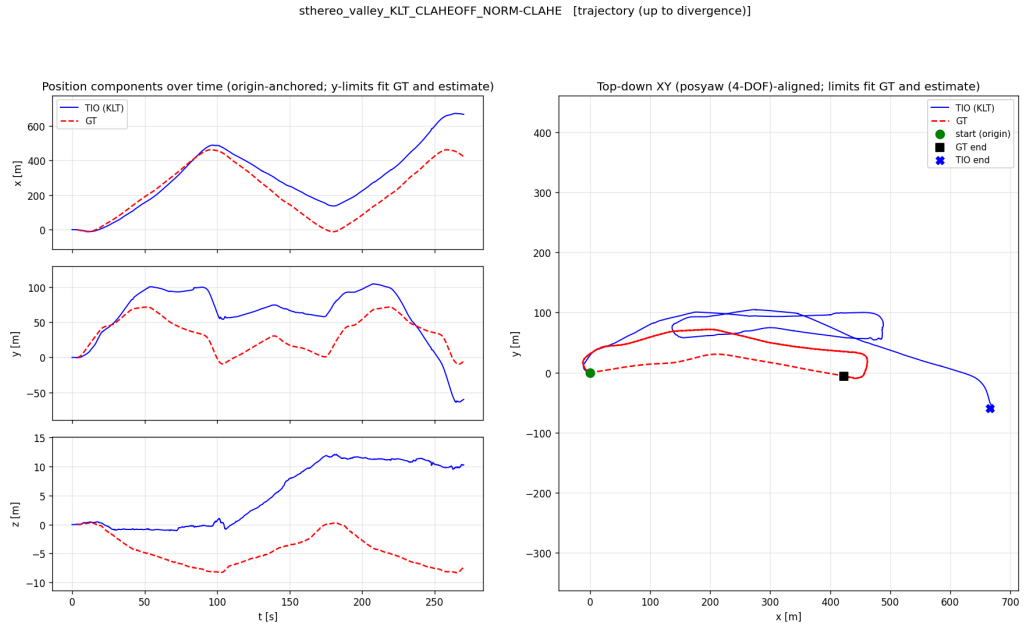
(a) Tracking window $[0, 120]$ s.

stereo_valley_ORB_CLAHEOFF_NORM-CLAHE [diagnostics]

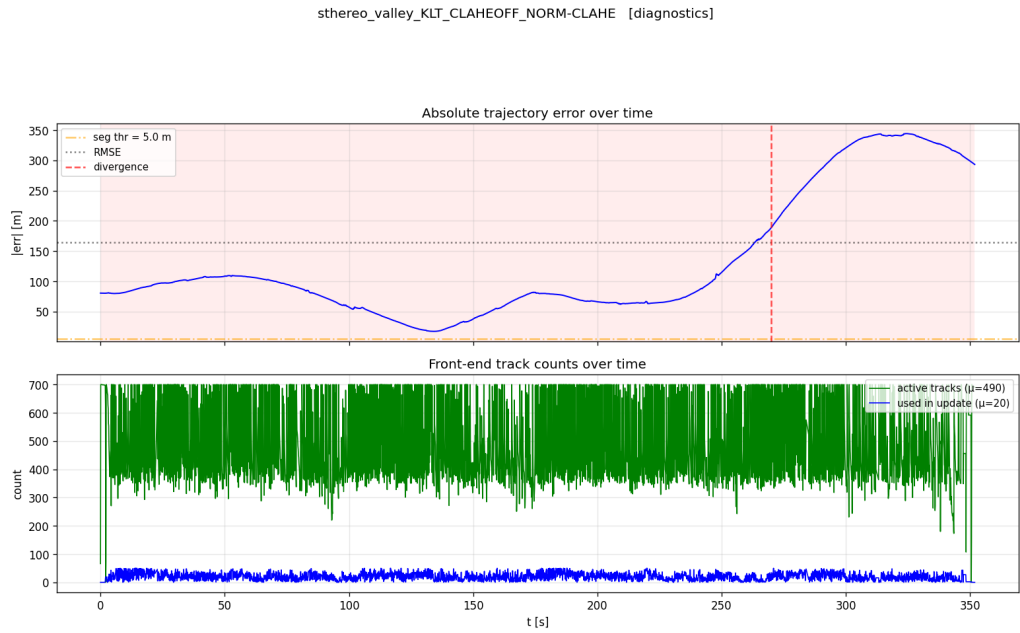


(b) Diagnostics: absolute position error and track counts over time.

Figure A.4: STheReo (ORB, CLAHE off): survives to $t = 120$ s, longer than the CLAHE-on ORB run of Figure 7.8 but still far short of KLT.



(a) Tracking window $[0, 270]$ s.



(b) Diagnostics: absolute position error and track counts over time.

Figure A.5: STheReo (KLT, CLAHE off): survives to $t = 270$ s at a somewhat higher RPE (1.11 m/m) than the CLAHE-on KLT run of Figure 7.6 (0.77 m/m), consistent with CLAHE mitigating the brightness-constancy violation of Section 7.6.

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